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Atmospheric water harvesting: A review of techniques, performance, renewable energy solutions, and feasibility

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ABSTRACT

Water scarcity is a major global challenge, with 2.3 billion people living in water-stressed countries and the potential for 700 million people to be displaced by 2030 due to widespread water scarcity. The increasing effects of climate change and overpopulation are exacerbating the problem, particularly in arid and remote regions. One potential solution is the harvesting of atmospheric water, which is estimated to be a vast source of freshwater at 12,900 km^3. This study aims to provide a comprehensive and up-to-date review of the latest research in the field of atmospheric water harvesting systems. The purpose of this review is to present the various types of harvested water, the use of solar energy, the methods of gathering it, and the mathematical models used in the literature. The paper has covered numerous types of atmospheric water harvesting techniques, including harvesting dew and fog. In addition, the paper has also discussed water harvesting technologies that rely on sorbent materials. The primary focus of the paper is to present recent advances in water harvesting systems, such as dehumidifying, condensing, vapor compression refrigeration cycles (VCRCs), thermoelectric coolers (TECs), air conditioning units, fuel cells, and integrated systems. The paper aims to provide a comprehensive understanding of the latest developments in atmospheric water harvesting systems. The paper also presents an assessment of the efficiency and effectiveness of these water harvesting systems. The paper concludes by comparing various approaches to provide accurate descriptive information regarding the amount of water harvested. Additionally, the article proposes suggestions to enhance current water harvesting systems and outlines potential future initiatives.

1. Introduction

The scarcity of water is considered one of the most pressing issues in the world. With an increasing population, the demand for water has also risen. This has led to a global challenge of water shortage, with many countries facing water deficiency due to factors such as climate change and overpopulation. Some regions, especially arid and remote areas, have become water-stressed zones as a result of global warming and other climate changes. Obtaining water for drinking and agriculture can be difficult in these regions.

There are attempts to address water shortage problems through projects such as seawater desalination and building more dams and wells, but these efforts may not be enough, particularly in dry and hot regions and areas far from water resources. According to the United Nations (UN-Water), 2.3 billion people live in water-stressed countries, and 733 million of them live in extremely and critically water-stressed countries. By 2030, it is estimated that around 700 million people could be displaced due to widespread water scarcity [1]. The Mediterranean region's atmospheric moisture levels are predicted to increase by 27% during the summer months by 2080, according to climate projections. These changes will boost the demand for non-traditional sources of fresh water, particularly atmospheric water [2,3].

The management of water resources is a crucial issue, particularly as many regions worldwide face problems such as water leakage through their supply networks and the depletion of water sources. In order to address these issues, various approaches have been suggested. One such approach is the use of a novel leakage detection model that applies spatial clustering, noise application, and multiscale fully convolutional networks to identify and prevent leaks in the water pipe system, thereby reducing water losses [4]. Another method involves using a sensitivity matrix of pipe flow to accurately detect water leakage points through pipelines, which can then be addressed to further minimize water losses [5]. In areas with oil and gas reservoirs, it is vital to determine the location of water in oil well production fluid due to water scarcity issues. To this end, a recognition model approach based on a machine learning classifier, the AdaBoost algorithm, has been proposed to identify

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Nomenclature

Abbreviations

AH	Absolute humidity
CFD	Computational fluid design
CIL	Composed of ionic liquids
COP	Coefficient of Performance
DC	Direct current
DBT	Dry bulb temperature, K
EES	Engineering equation solver
GRR	Global Risk Report
HVAC	Heating, ventilation, and air-conditioning
IoT	Internet of thing
MCDM	Multi Criteria Decision Making
MHI	Moisture harvesting index
MOFs	Metal-organic frameworks
PV	Photovoltaic
RH	Relative humidity
TEC	Thermoelectric coolers
UN	United Nations
UPC	Unit power consumption
USD	United States dollar
VCRC	Vapor compression refrigeration cycle
WBT	Wet bulb temperature, K
WH	Water harvesting
WHR	Water harvesting rate
WHE	Water harvesting efficiency

List of symbols

A_{FC}	The fuel cell area, m^2
C_p	Daily water demand, m^3
D	Real-time amount of water vapor
E_{eq}	Device effectiveness
E_g	Latent heat of the water
F	Faraday's constant
h	Specific enthalpy, kJ/kg
h_{fg}	Latent heat of condensation, kJ/kg
j	Fuel cell current density
M_w	Molecular weight$\dot{m}_a$Mass flow rate of air, kg.s^{-1}$\dot{m}_w$Mass flow rate of water, kg.s^{-1}N Molecular weight$\dot{m}_a$Mass flow rate of air, kg.s^{-1}$\dot{m}_w$Mass flow rate of water, kg.s^{-1}NPopulation
P	Atmospheric pressure, Pa
P_s	Saturated water vapor pressure
Ƥ	Power per unit area W/m^2
$\dot{Q}_{cool}$	radiative cooling, W
S_t	Mesh area, m^2
T	Temperature, K
T_a	Ambient air temperature, KT_d Ambient air temperature, KT_dDew point temperature, K
T_s	The surface temperature, K
U_z	Wind speed at specific height, m.s^{-1}
V_c	Water volume per area, m^3/m^2
$\dot{W}$	Work, W
η	Efficiency
ρ_w	Density, kg/m^3
ε	The surface emissivityσ The surface emissivityσThe Stefan-Boltzmann constant which is $5.67x10^{-8}$ $W/m^2.k^4$
ω_{cap}	The mass of the captured water

water-flooded layers [6]. The severity of water scarcity worldwide is reflected in the fact that half a billion people experience it all year round [7]. The Global Risk Report (GRR) by the World Economic Forum has also highlighted water crises as one of the top-five risks in terms of impact for eight consecutive years [8].

Despite the challenges posed by water scarcity, there is a large enough amount of water in the atmosphere that can help reduce its effects. This atmospheric water is estimated to be 12900 km^3. There are studies aimed at finding ways to extract atmospheric water, such as using sorbent and desiccant materials, and vapor compression cycles.

Renewable energy sources like solar photovoltaic (PV) systems, wind energy, and biomass have the potential to play a crucial role in atmospheric water harvesting. These sources can power water generation systems like solar-regenerated desiccant systems, which use heat and desiccant materials to extract moisture from the air. Solar chimneys and wind turbines can also be used to create the conditions necessary for water condensation and collection. Various review papers in the literature have focused on different aspects of atmospheric water harvesting, such as materials used in research or specific technologies like metal-organic frameworks (MOFs). Researchers like Pan et al. [9], Feng et al. [10], and Parvazinia [11] have conducted reviews on topics ranging from MOFs to solid or liquid desiccant materials. However, our aim was to provide a comprehensive overview of all techniques and methods used in atmospheric water harvesting in one paper, including comparisons of the amount of water collected. To achieve this, we developed new schematics and conducted an in-depth analysis of all procedures involved in atmospheric water harvesting.

This paper provides a comprehensive review of the latest research in the field of atmospheric water harvesting systems. The review covers various types of harvested water, methods of gathering it, and mathematical models used in the literature. It also discusses recent advances in water harvesting systems, such as dehumidifying, condensing, VCRC, TEC, air conditioning units, fuel cells, and integrated systems. The review will assess the efficiency and effectiveness of these water harvesting systems.

The review will focus on several aspects of atmospheric water harvesting, including different types of systems, their performance and usability. This includes a close examination of fog harvesting, where water is collected from the atmosphere using specially designed structures, and dew harvesting, where water condensation from the air onto surfaces is captured. The review will also present mathematical and thermodynamic models used to understand and optimize the performance of these systems. It will also examine advances in water harvesting systems, including the use of renewable energy and the latest technology and design developments, and the incorporation of new materials and methods to improve efficiency and reduce costs. The overall aim of the review is to provide a comprehensive overview of the current state of the art in atmospheric water harvesting and identify areas for future research and development. By examining different types of systems, models, advances, and assessments, the review will provide valuable insights into the potential of atmospheric water harvesting as a sustainable source of water for communities in arid and semi-arid regions. Finally, the review will consider the progress made in the assessment of water harvesting systems, including new methods for measuring performance, challenges and limitations, and possible solutions.

2. Atmospheric water harvesting types

Water is vital to all forms of life, but due to factors such as population growth, some communities are experiencing water scarcity. This has led to the exploration of alternative sources of water, including those found in the atmosphere. Fog and dew are two types of atmospheric water that can be harvested using specialized systems, and they have the potential to provide freshwater in regions facing water scarcity. To design effective atmospheric water harvesting systems, it is important to understand the different types of harvestable atmospheric water and the conditions under which they can be collected. This article provides an overview of the principles and developments in the field of harvesting dew and fog atmospheric water.

2.1. Fog harvesting

In some regions, fog harvesting is a beneficial source of water, but it is limited by the occurrence of fog. In contrast, dew-gathering technology can be used wherever a cooled surface is available. Atmospheric water harvesting systems aim to provide water regardless of complexity, humidity level, or location. Fog collectors consist of a small mesh that captures water droplets from the air and gravity causes them to fall to the ground [12].

The fog collector is based on a straightforward and conventional idea; mesh is set up in a foggy area and left exposed to the wind. After fog hits the mesh, the water molecules in the fog begin to drip down within a collector tank due to gravity. In Fig. 1, a basic fog machine is shown. There are two types of fog collectors: standard fog collectors and large fog collectors [13]. Large fog collectors are utilized for large water yields, whereas standard fog collectors are employed for small-scale applications. The occurrence of global fog is a key factor in fog water harvesting [12].

The feasibility of harvesting fog and rain was studied in Nepal [14] and Syria [15] to address water scarcity. In Nepal, a polypropylene mesh fog collector and roof system were used to collect 2.5 m^3/m^2 and 1.5 m^3/m^2 of water respectively, providing 1000 L of drinkable water meeting quality standards. In Syria, two fog collectors were built and coated with specific materials to increase efficiency. It was concluded that the mesh material and design geometry played a critical role in enhancing the condensation process. It is important to note that absorbent materials can absorb water vapor at any temperature [10]. Both studies indicate the potential of atmospheric water harvesting systems as an additional water source in areas facing water scarcity. An experimental and model study was conducted to examine various parameters that affect the condensation process of air moisture at the mesh surface. The experimental findings showed that the mesh material and design geometry had played a critical role in enhancing the condensation process [16].

Given the significance of water scarcity, it is important to highlight the potential of fog and atmospheric water harvesting as additional sources of clean drinking water. These techniques have been shown to be effective in harvesting substantial amounts of water, with their success relying on factors such as surrounding air quality, mesh material, and fog collector design geometry. Ongoing research has demonstrated that improvements in design and materials can further enhance the efficiency of these techniques. Thus, it is imperative to recognize the potential of fog and atmospheric water harvesting as an alternative to conventional water sources in areas with limited access to clean drinking water.

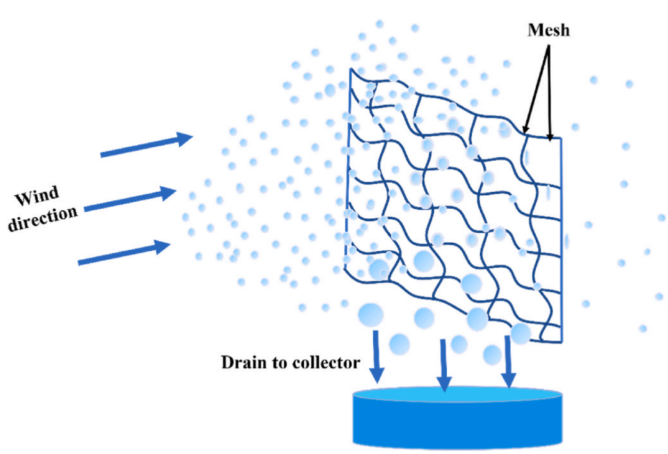

Fig. 1. Basic fog machine.

2.2. Dew harvesting

Because water vapor is ubiquitous in the atmosphere, condensation can be employed to harvest water. Dew water is water that condenses on a surface once the air temperature is lowered below its dew point temperature. Dew water harvesting can be divided into three main categories: passive or radiative cooling condensers approach, water generation via a solar-regenerated desiccant system, and water collection via an active cooling condensation method [12]. It is critical to note that natural dew formation is mainly influenced and related to environmental factors and phenomena such as temperature, relative humidity, airspeed, cloud cover, and material properties [17], Furthermore, the amount of harvested water in active water collecting devices is directly proportional to the amount of input energy [18]. The dew-gathering process is dependent on natural cooling power, which is usually between 25 and 100 W/m^2 and is limited to a dew yield of under 0.8 $L/m^2/day$ [19].

A study, conducted in Iran [20], identified optimal dew harvesting locations based on relative humidity, wind speed, and sunny days. Regions with over 70% humidity, low wind speed (<3 m/s), and high sunshine were ideal. Around 17% of Iran's locations were found suitable. Researchers have improved efficiency with hydrophilic micro-patterns [21], silicon-based water panels [22], and wettable surfaces. Fog collision with surfaces is the primary mechanism for fog harvesting [23]."

In Abu-Dhabi, a prototype square funnel made of aluminum was developed and tested for water harvesting from dew, fog, and rain. The findings showed that dew occurs 100 days out of the year, with an average of 0.016 ml/day (0.016 $mm/m^2/day$) of collected water [24]. These studies suggest that dew harvesting has the potential to be an additional water source in areas affected by water scarcity, and engineers are exploring different ways to optimize the process for maximum efficiency [25].

Dew harvesting is a promising and vital solution for water scarcity. Regions with high humidity, low wind, and sunny days are ideal. Experimental studies show enhancements in efficiency using hydrophilic patterns, silicon panels, and wettable surfaces. Engineers continue to optimize dew harvesting for maximal water collection, demonstrating its potential as an additional water source.

2.3. The radiative cooling

Radiative cooling is a technique that uses the natural cooling ability of the Earth's surface to harvest atmospheric water. The process involves exposing certain materials to the sky, allowing them to cool below the ambient temperature and condense atmospheric water vapor. The Earth's surface cools down by emitting thermal radiation, which is absorbed by the atmosphere and radiated back to space. Radiative cooling can lower surface temperatures below the dew point, resulting in the condensation of atmospheric water vapor and the creation of harvestable water. This method shows promise for water harvesting in arid and semi-arid regions, where traditional methods may be less effective. It is a simple process that depends on the power gradient between sky radiation and condenser outgoing radiation power, as described by the Stefan Boltzmann law as follows [12]:

$$P = \varepsilon \sigma T_s^{\,4} \tag{1}$$

where:

P: the radiative power per unit area W/m^2A1

ε: the surface emissivity

σ: the Stefan-Boltzmann constant which is $5.67x10^{-8}\ W/m^2.k^4$A1
T_s: the surface temperature

Chronologically, the effect of condensing surfaces fabricated from different materials has been studied under Bahrain climate conditions, it was observed that the condenser surface that was made of aluminum had the largest dew collected compared with other than glass and polyethylene foils surfaces [26]. Within this content, it is also critical to discuss and emphasize the widespread use of various types of absorbents with various approaches to harvesting water molecules. Fig. 2 depicts the concept of passive water harvesting systems [27].

Adsorption-based atmospheric water harvesting methods have the advantage of being able to collect and obtain water at low relative humidity values. Zeolite is the first adsorbent used in such systems, and there are several types of adsorption-based atmospheric water harvesting materials with different structures [28]. These include physical adsorbents like silica gel, polymeric adsorbents such as metal organic frameworks, and chemical sorbents like hygroscopic salts [29]. To design highly efficient sorbents, it's important to explore the interaction between water and sorbent materials and investigate mass and heat transfer effects. The sorbent materials are classified into four groups: nano porous solids, hygroscopic polymers, salt-based composites, and liquid sorbents [30]. Adsorption-based atmospheric water harvesting devices can be either active or passive, depending on the energy source [31]. However, the major weakness of the adsorption-based water harvesting systems is that they cannot run continuously [32].

This section covers the radiative cooling mechanism and highlights the significance of the adsorption-based method for atmospheric water harvesting. In the following section, we provide a detailed overview of several key studies to further enhance comprehension of this phenomenon.

2.4. Applications and developments using radiative cooling

A theoretical framework was built and studied to analyze the effect of some parameters such as ambient temperature, relative humidity, and convection heat transfer coefficient on dew harvesting technology via radiation cooling. The surface of the condenser is radiatively cooled below its dew point temperature, causing adjacent atmospheric water molecules to condensate and collect. The effect of convection phenomena has been studied and was found in two paths. It was observed renewing in the freshwater vapor at the condenser surface to gain more condensed water throughout the convection effect, while the bad side effect of the convection was observed since it warms the condenser surface. Therefore, optimization of the convection coefficient is required to yield the maximum amount of harvested water. The effect of the convection process is shown in Fig. 3 [33].

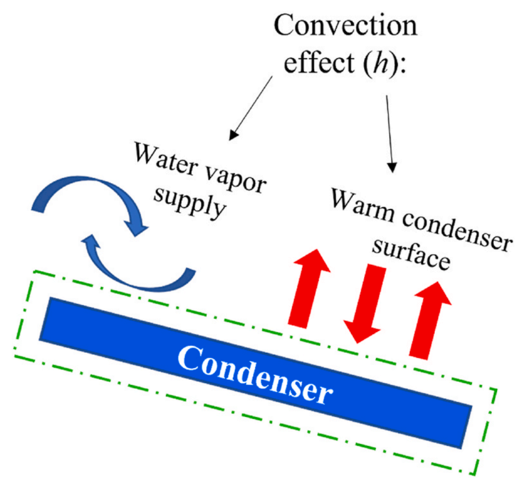

Fig. 3. Influence of convection process.

Various studies have investigated the potential of radiative cooling for water harvesting. One study found that using a black emitter, the maximum water that could be harvested was 60 g for 2 h at 10% relative humidity [33]. Another study conducted in Dubai found that solar panel surfaces can produce an average of 261 ml/m^2 of water per week [34]. In Brazil, an experimental study compared different condensing surfaces and found that a standard plastic surface collected the most water [35], but a new cone-shaped device designed solely for radiative cooling was able to collect condensed water [36]. Finally, a solar-powered interfacial evaporation apparatus that combined solar-powered interfacial desalination with dew water generation was able to produce clean drinking water with an efficiency of 71.1% [37].

As mentioned earlier, radiative cooling has been explored in several studies, and its effectiveness largely depends on the radiative properties of the surface. Although solar cells and other engineered surfaces have been utilized to collect atmospheric water through this approach, the quantity of water collected remains insufficient and requires improvement to achieve optimal performance.

2.5. Water generation via solar-regenerated desiccant systems

Water is generated by desiccant materials like zeolites hygroscopic and silica-gel to absorb air moisture during night period and to regenerate water by desorption process during day times. The desorption process includes the releasing of water molecules from desiccant materials through heating by solar radiation, but one of biggest limitations

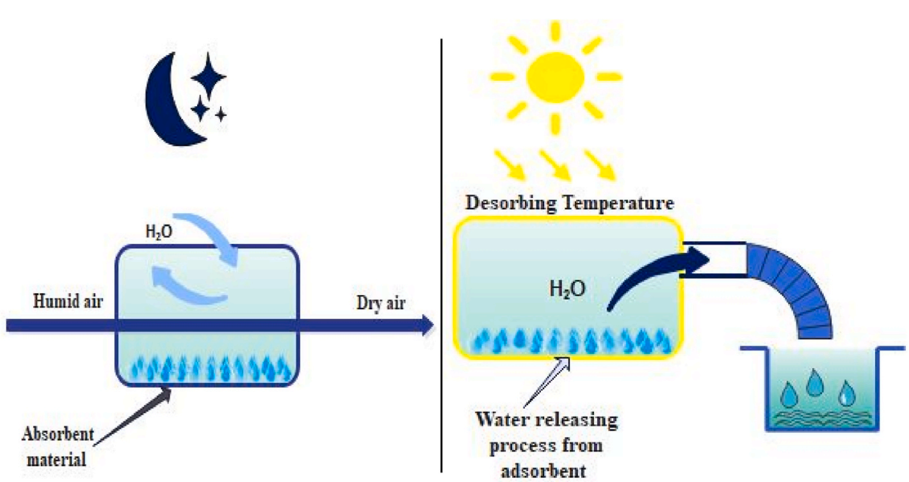

Fig. 2. Passive extraction of water from air during day and night using desiccant substances.

and drawbacks of this method that is its low water yield [12]. Adsorption/absorption-desorption systems with desiccant material can frequently achieve a specific water yield of 0.1–1 L/kWh [27]. The idea of the water harvesting technique utilizing desiccant materials is illustrated in Fig. 4.

Several experiments have been conducted to explore the use of air-cooled adsorption-based devices for atmospheric water harvesting with different materials. These experiments include using activated carbon fiber felt-silica solution-LiCl30 to collect 7.7 kg of water per cycle [38], a new sorbent material called LiCl@MIL-101(Cr) that harvested water with a capacity of 0.77 g/g [39], and a silica gel model that collected 159 g of water per cycle [40]. Additionally, various methods have been proposed to harvest atmospheric water using different materials and techniques, including a sorption material device that collected 0.25 L of water per kilogram of metal-organic framework material per cycle [41], a dual-stage device that used zeolite material and sunlight to harvest water at a higher rate [42], and a device that used a new sorption material with high salt content and a membrane to prevent leakage and collected 560 ml of water in an outdoor experiment [43]. Lastly, two solar-based atmospheric water generators were manufactured and evaluated, with water productivity affected by factors such as the amount of sorbent, heat from the solar collector, and the sorbents' mass transfer capacity [44].

A multistage desiccant wheel was proposed to moisten the air. The proposed system consists of multistage desiccant wheels, an adsorption process, and a heat pump system. The air was humidified using wheels while the role of the heat pump system was used to raise the temperature of the regeneration air and to condense the humidified air to produce water. After passing through desiccant wheels, the moisture in the air is adsorbed and dehumidified, and the dehumidified air is regenerated through ambient air to become humid regenerated air. The desiccant wheels are used to absorb water vapor ambient air and transport it to the regenerated air. Following that, a vapor compression cycle is used to condense water vapor from the humid air, in which the suggested system can offer 32.5 kg/h [45]. A multistage desiccant wheel system was proposed as a water generator that relies on heating and cooling processes. The optimal number of stages, structure design, and configurations for the system were determined through theoretical analysis. It was recommended to increase the stage number and utilize 100 mm as the thickness of the used desiccant wheel for each stage to increase the harvested amount of water. With parallel configurations of three humidification units at an ambient temperature of 28 °C and a 20 g/kg humidity ratio, a maximum water gathering rate of 95.2 kg/h was achieved [46].

Several different systems for harvesting atmospheric water using solar energy were studied. One system uses a solar tabular still in low humidity regions in Saudi Arabia, which collects about half a liter of water per kilogram of desiccant material, as shown in Fig. 5 [47]. Absorption mechanism can be classified as a bulk phenomenon wherein gas or liquid molecules diffuse into liquid/solid materials. However, the ideal harvesting materials must have porous building to maintain rapid vapor diffusion during capturing air moisture, suitable interaction with the water molecules for easy releasing, and stable construction [48]. A second system is a quasi-continuous system powered by solar energy that uses a microporous aluminum-based metal-organic framework (MOF-303) that can complete the adsorption-desorption cycle in minutes. This system was able to produce 1.3 L of water per kilogram of adsorbent per day in an indoor experiment and 0.7 L of water per kilogram of adsorbent in the Mojave desert [49].

The design and testing of various materials and devices for absorbing water vapor from the air and converting it into liquid water were discussed. These include a membrane hydroscopic composite nanofibrous material that can absorb 0.96 g of water per gram of material at 30% relative humidity [50], a graphene oxide aerogel-based atmospheric water generator that uses solar energy to generate 2.89 kg/m^2 of water per day at 70% relative humidity [51], and a semi-open atmospheric

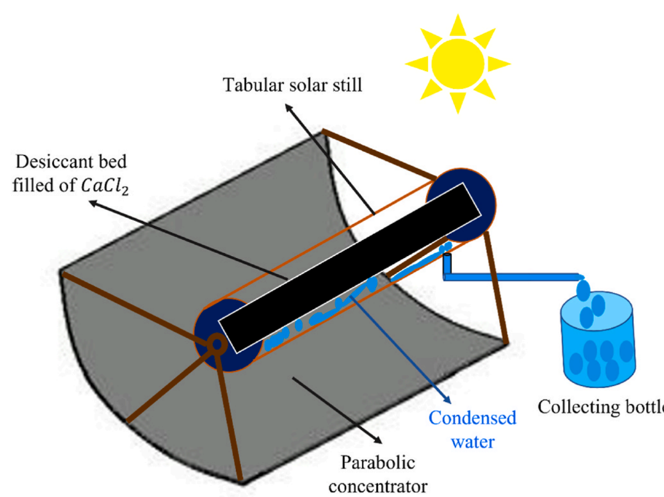

Fig. 5. Tabular solar still for atmospheric water harvesting.

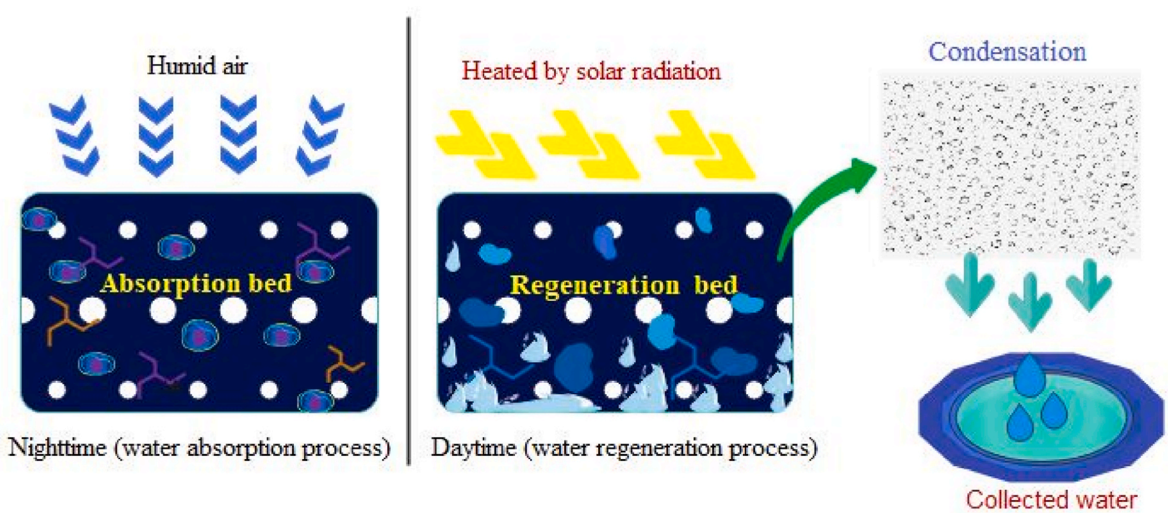

Fig. 4. Water production process from atmospheric air using wet desiccants.

device that uses activated carbon felt incorporated with lithium chloride as a fast and high-performance sorbent material [52]. Another article also mentions research on the use of composite materials containing lithium chloride and magnesium sulfate to enhance the efficiency of the water absorption process and reduce the risk of leakage, a prototype had been fabricated utilizing the composite materials and powered by solar radiation energy. It has shown high performance at 35% relative humidity surroundings and satisfies 0.92 $g_{water}/g_{material}$ [53].

The design and testing of various devices for absorbing water vapor from the air and converting it into liquid water, specifically in the context of Saudi Arabia was discussed. One device is a trapezoidal prism filled with black cotton rag saturated in calcium chloride, which can yield 1.06 L/m^2 of water per day. The developed model of trapezoidal prism for atmospheric water harvesting is illustrated in Fig. 6, wherein it is intended to absorb air humidity at night and offer water during the day [54]. Another device uses silica gel as a sorbent material to trap and capture water molecules from the air, but it performed poorly due to experimental error [55]. A liter-scale atmospheric water harvesting device using an adsorption heat exchanger based on silica gel as a sorbent material was also designed and investigated, which could harvest between 1.5 and 3.3 L/day per square meter of solar field when the ambient and regeneration temperatures were 35 and 57 °C, respectively. The article also mentioned the drawback of water harvesting devices that were working based on sorbent materials was required a large surface area to provide massive water [56].

The field of atmospheric water harvesting has seen significant progress in recent years with a focus on improving the efficiency and effectiveness of water collection. A variety of methods and materials have been studied and tested.

To further emphasize this point and provide more clarity on the topic, the following highlights have been divided into four subsections that showcase various materials utilized in water harvesting systems. Additionally, a summary table (Table 1) has been created for ease of reference.

2.5.1. Some sorbents like cottons

The process of water harvesting has been achieved using cotton rod fibers. An adsorption-based device was designed and fabricated using a cotton rod fiber, which was composed of ionic liquids (CIL) and carbon nanotubes to capture water molecules and convert light to heat, respectively. This device was able to harvest atmospheric water, even in temperatures below 10 °C, by initiating the solar evaporation process. In an outdoor experiment conducted under low humidity (30%), less than 10 °C temperature, and radiation flux of 0.3–0.8 kW/m^2, the device was able to harvest 1.49 kg/m^2 of water per day [58]. Another study focused on the impact of the sorbent material on the system-level performance of atmospheric water harvesting systems. The equilibrium properties of the material were found to be a good indicator of system performance, and understanding the mass and heat mechanism of the material is crucial in developing new sorbent materials. The sorption mechanism is affected by various factors, including vapor diffusion through the adsorbent structure, which is described by Fick's law of diffusion. The material properties, such as pore volume and adsorption enthalpy, and component properties, such as vapor transport to the condenser and layer thickness, also affect the sorption mechanism [63]. To determine the size of the atmospheric water harvesting system, it is suggested to express daily water production as a function of the area exposed to the heat source.

Table 1
Summary of recent materials used to harvest atmospheric water.

Ref.	Procedure	Sorbent Materials	Amount of harvested water
[38]	Experimental	21 kg ACFF-silica solution-$LiCl_3$	7.7 kg per day and night cycle @63% RH
[39]	Experimental	LiCl@MIL-101(Cr)	0.45 $kg_{water}/kg_{sorbent}$ @30% RH
[40]	Experimental	Silica gel	159 $g_{water}/kg_{sorbent}$ in 12 h
[41]	Experimental	MOF-801 $[Zr_6O_4(OH)_4(furmarate)_6]$	0.25 L/kg_{MOF}
[42]	Experimental	Commercial zeolite (AQSOAZO1)	0.77 $L/m^2/day$
[43]	Experimental	High salt content composite (HSCC-E10)	560 mL/m^2 in 4 h through a lab scale device 3.75 $g_{water}/g_{sorbent}$ @90% RH
[57]	Theoretical and experimental	Nonporous super hygroscopic hydrogel	10 L/kg
[49]	Experimental	Microporous aluminum-based MOF-303	1.3 L/kg_{MOF} at indoor experiment @32% RH
[49]	Experimental	Microporous aluminum-based MOF-303	0.7 L/kg_{MOF} at Mojave Desert @10% RH
[50]	Experimental	Hygroscopic LiCl incorporated in graphene oxide-Polye (VCL-AA0 composite hydrogel nanofibers.	0.96 $g_{water}/g_{sorbent}$
[51]	Experimental	calcium chloride with graphene oxide GO aerogel	2.89 $kg/m^2/day$ @70% RH
[53]	Experimental	ACF containing LiCl and $MgSO_4$	0.92 $g_{water}/g_{sorbent}$ @ 35% RH
[54]	Experimental	Black cotton cloth bed saturated in chemical solution ($CaCl_2$)	1.06 $L/m^2/day$ @26.5% RH
[58]	Experimental	Composited ionic liquid (CIL) with carbon nanotubes photothermal materials	1.49 $kg/m^2/day$ @30% RH
[59]	Experimental	MOF-801/G MOF-803/G	100 $g_{water}/kg_{sorbent}/day$ 174 $g_{water}/kg_{sorbent}/day$ @RH = 30–50%
[60]	Experimental	Desiccant material containing $CaCl_2$ in concentration of 30%	2.320 $L/m^2/day$ for cloth bed 1.235 $L/m^2/day$ for sand bed
[61]	Experimental	Desiccant bed impregnated with 30% concentration of $CaCl_2$	1.01 $L/m^2/day$ @NA RH
[62]	Experimental	Nanocarbon hollow capsule with LiCl	1.6 $kg_{water}/kg_{sorbent}$ during 10 h @60% RH

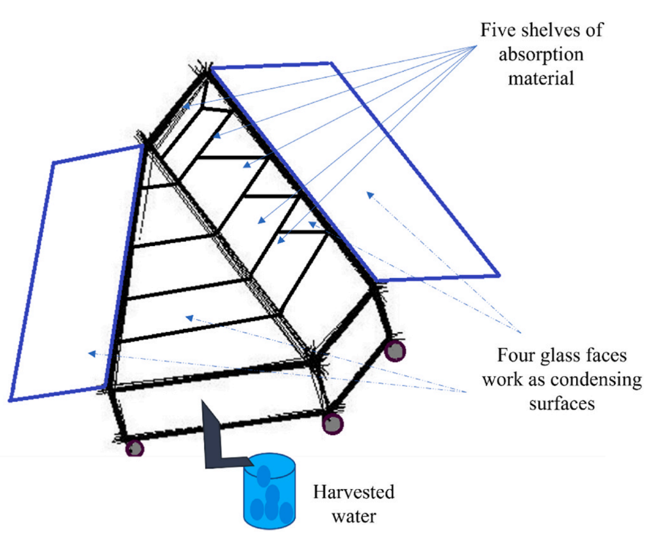

Fig. 6. Trapezoidal prism atmospheric water harvesting model.

2.5.2. Metal-organic frameworks (MOFs)

Metal-organic frameworks (MOFs) materials are frequently used for capturing water, but despite their benefits, most suffer from low water yield. A study proposed four green approaches for synthesizing MOF-303 quickly to reduce production time [64]. An experimental test was conducted in Arizona's desert using a MOF-801 prototype, which harvested 100 g of water per 1 kg of MOF-801 daily without an external cooling source and using aluminum-based MOF-303 was found to produce more than twice the harvested water. To examine how temperature sources and adsorbent material options affect water harvesting efficiency, materials such as MOF-801 and MOF-303 were investigated [59]. For example, at relative humidity values below 40%, they exhibited thermal energy-to-water conversion efficiency and second law efficiency of 0.33 and 0.18, respectively. Additionally, a cascading concept was introduced to efficiently run two adsorbent beds in sequence, with higher values of thermal and second law efficiencies observed [65].

2.5.3. Desiccant materials

Numerous studies have been carried out on the use of desiccant materials, which can be categorized as liquid or solid. Recently, a review has been published that focuses on the advancements made in using solid desiccant materials as water harvesting tools [11]. As part of this research, a trapezoidal prism solar collector was developed and built with a multi-shelf bed to increase the surface area that collects atmospheric water. Desiccant materials containing a $CaCl_2$ solution were used to capture water vapor from humid air and produce water. A cloth bed was used to obtain a total of 910 g/m^2. day of water during June [60]. A sand bed saturated with calcium chloride was also created and tested in Taif, Saudi Arabia. A water extraction device was fabricated, and it had a surface area of 0.5 m^2, working based on desiccants. The water absorption process occurred at night, and during sunshine periods, the water generation process began while the desiccant bed was regenerated. The water harvesting unit was capable of producing 1.01 L/m^2 in Taif's climate conditions and with a 30% concentration of calcium chloride solution [61]. In addition, a theoretical and experimental study was conducted on a foldable atmospheric water device to harvest atmospheric water under the Mansoura region's climate conditions in Egypt. An absorber saturated with $CaCl_2$ was used to capture air humidity at night, and the water molecules were evaporated and condensed in a bottle with the help of solar radiation during the day. The design and analysis of the device were carried out using SolidWorks and MATLAB software. It was found that the acquisition and harvesting of air moisture depended on the partial water vapor pressure difference in the device. However, the highest recorded efficiency reached 30.6%, while the maximum water production was 750 g/day [66].

2.5.4. Some super gel materials

Super gel materials had been investigated and created for examples; in a distinguished work, super gel material had been synthesized, it has two states: hydrated and dehydrated, and it was used for energy transformation and water harvesting. The fabricated dehydrated gel had been exposed to the ambient air and for 2 h, the dehydrated structure of gel had absorbed a high amount of water may reach 200% by its weight in conditions of 90% relative humidity. After the dehydrated gel absorbed air moisture, it had become hydrated and through exposing it to solar energy, water is generated. Furthermore, the synthesized gel had a wide range of applications such as: energy harvested by using it to run electromechanical cell [67].

A creative sorbent was developed to achieve more water production cycles. Nanocarbon hollow capsule with LiCl inside its void core. The nano sorbent that was fabricated can capture the air moisture with high efficiency and may reach 100% of its weight during 3 h when the air relative humidity is 60% and it can release the water from the sorbent in half hour when it is exposed to 1 kW/ m^2 of sunlight irradiation so it is considered quick water production materials. A prototype was designed and successfully tested the proposed sorbent. It is concluded that it can complete 3 water production cycles within 10 h and release 1.6 kg of water per 1 kg sorbent [62]. Another system uses a hydrogel with a nano-pored structure and super hygroscopic properties, which can absorb over 420% of its weight in water [57]. Finally, an atmospheric water harvester based on mechanically biophilic sorption had been investigated. Analyses show that at 97% relative humidity and 25 °C, the produced sorbent can capture 86.3 g of water per gram of sorbent for one day and that 88% of sorbent material can be recovered within 150 s of mechanical squeezing. It is a promising material, but further work is needed to boost its capacity for water collection [68].

2.5.5. Salts

Also, the salts have been used to capture air moisture in different atmospheric water devices as followings: a mathematical model was introduced to simulate and model the absorption process of three different anhydrous salts which are copper chloride, copper sulfate, and magnesium sulfate, the authors compared the computed model with an experimental result. Anhydrous salts can be considered an excellent choice in many regions in which relative humidity is around 10%, furthermore, these salts just crave any thermal source to start the water generation process. Among the three salts, copper chloride was having the highest saturation rate whereas the relative humidity was 15% [69]. An atmospheric water harvesting apparatus based on a high concentration watery salt solution of lithium bromide has been developed but requires pressure exchangers and a modified membrane for reverse osmosis when the absorbents have low vapor pressure [70].

In conclusion, the reviewed studies have demonstrated that atmospheric water can be collected through diverse techniques, and certain methods exhibit higher efficiency rates than others. Materials such as ionic liquids, carbon nanotubes, and metal-organic frameworks have demonstrated promising results in water harvesting systems. However, some techniques necessitate specific conditions or components, while others have the potential to be utilized in areas with low humidity levels. Despite these advancements, the field of atmospheric water harvesting still has plenty of room for improvement, and further research is essential to advance the understanding and optimization of these techniques.

2.6. Water collection using active cooling condensate method

In fact, active cooling for atmospheric water harvesting requires electrical power to run a compressor or vacuum pump, and it has a high-water yield if an active harvesting device was indeed continuously used [12]. Because dew water harvesting technology is not affected by climatic conditions, it can be considered an alternative available technology that can provide water to the most environmentally sensitive areas. Furthermore, the amount of harvested water in active water collecting devices is directly proportional to the amount of input energy [18] so how much energy it can provide reflects how long it will gain atmospheric harvested water. Most of the systems operate on the evaporation-condensation principle, and most active water harvesting systems are designed using a vapor compression cycle. However, active water harvesting devices may be designed using desiccant wheels. Fig. 7 depicts two approaches generally followed in designing and constructing active water harvesting cycles [27].

The commercial active water harvesting devices typically yield 2–4.2 L/kWh [27,71]. A world map was created based on climatic data that considered annual average relative humidity and temperature during the night. The map showed that active air-cooling systems are effective for harvesting atmospheric water in low relative humidity, high-temperature regions like deserts. In contrast, desiccant systems are recommended for capturing airborne water in low-temperature and very low relative humidity areas. Regions like North Africa, the Middle East, and Australia, where relative humidity ranges from 20% to 40% and temperature ranges from 20 to 30 °C, require specific attention. Active air-cooling systems and desiccant-based technologies are the preferred methods for harvesting atmospheric water, with active air-cooling being more effective in high-relative humidity areas of 40–60% and temperatures ranging from 0 to 20 °C [72].

In conclusion, Active water harvesting technologies are based on the use of energy to operate specific prototypes or devices, which can be

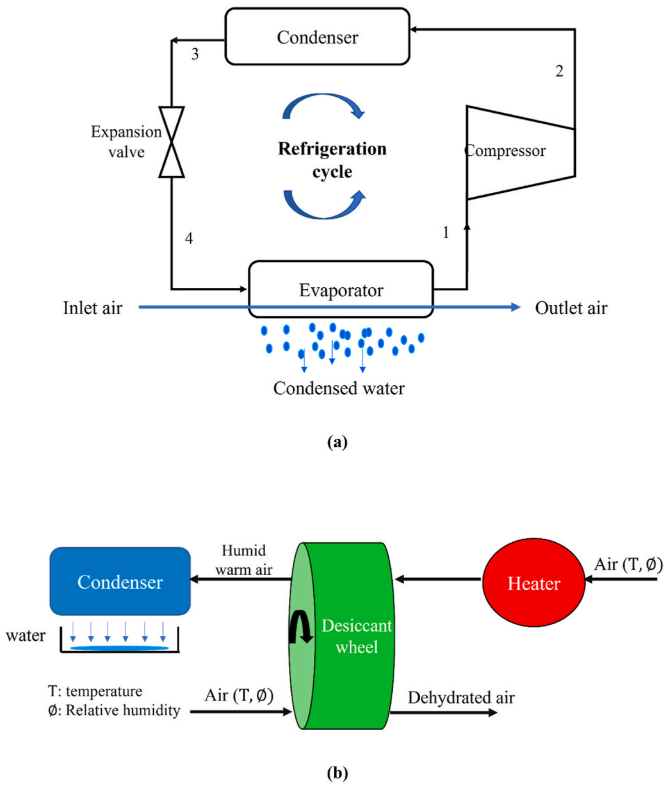

Fig. 7. Active water harvesting machines: (a) using vapor compression cycle, (b) using desiccant wheels.

powered by sorbent materials or cooling the air below its dew point temperature. These technologies are capable of collecting significantly more water compared to passive methods.

$$D_y = \begin{cases} \left\{\left[0.37\left(1+0.204323H-0.0238893H^2-\{180123-104963H+0.21891H^2\}10^{-3}T_d\right)\left(\frac{T_d+273.15}{285}\right)^4 x\left(1-\frac{N}{8}\right)\right\} \\ \quad +[0.06(T_d-T_a)]\left(1+100\left\{1-e^{\left(-\left(\frac{s}{s_o}\right)^{20}\right)}\right\}\right) & \textit{if positive} \\ 0 & \textit{if negative} \end{cases} \tag{7}$$

3. Mathematical and thermodynamic modeling

In this section, several studies are reviewed and stated to illustrate the main mathematical models and methods that have been presented in the water harvesting field.

To assess the fog and rain collection, a methodology was utilized based on observations that were collected and made at the project site furthermore, a mathematical relation had been used [14]. The required mesh area was designed and calculated as follows in Eqn. (2), while considering the following parameters: population that the device wanted to supply (N), the daily water demand (C_p), as well as average volume of the produced water in the study area (V_c).

$$S_t = \left(N\ x\ C_p\right) / V_c \tag{2}$$

A methodology had been started and developed based on Multi Criteria Decision Making (MCDM), [16]. Two model were proposed to harvest fog water [15]. The authors had stated that to calculate the water harvesting rate, Eqns. (3)–(5) were used. Eqn. (3) was used to calculate the dew point temperature, and the relation between air content of water, dew point temperature, and the absolute humidity (AH) was given by Eqn. (4). Accordingly, to determine the amount of accumulated water in liters in one day or 24 h Eqn. (5) was used to accommodate the device effectiveness E_{eq} and the wind velocity at specific height U_z.

$$AH = \frac{6.112\ x\ e^{\left[\frac{17.67\ x\ T}{T+243.5}\right]}\ x\ RH\ x\ 2.1674}{273.15+T} \tag{3}$$

$$T_d = T_a - (RH+100)/5 \tag{4}$$

$$WH = 24\ x\ AH\ xU_z x\ E_{eq}\ x\ 3.6 \tag{5}$$

Where T: air temperature, T_d: dew point temperature. T_a: ambient air temperature, RH: relative humidity.

The solar powered water harvesting devices were investigated based on desorption and sorption processes, the desorption phase process has plotted on the psychrometric chart [44]. The air temperatures and relative humidity's were recorded by a specialized instrument. After experimental studies were done the real-time amount of water vapor is determined as following Eqn. (6):

$$D = \frac{0.622\ x\ \varphi\ x\ P_s\ x\ m}{(P-\varphi\ x\ P_s)} \tag{6}$$

Where D: real-time amount of water vapor, m: mass flowrate, P: the atmospheric pressure, P_s: saturated water vapor pressure.

To analyze different condensing surfaces, experimental work had been carried out an the data was analyzed [35]. Eqn. (7) could be used to estimate the dew yield. Furthermore, the authors had referred to Eqn. (8) in which the proportionality factor α was given and stated the relation between estimated collected water theoretically and experimentally.

$$\sum_{d_f}^{d_1} D_{ye} = \alpha \sum_{d_f}^{d_1} D_{yc} \tag{8}$$

Where Dy: dew yield during the night, considering 12 h (mm/night), H: Site elevation (km), N: cloud cover (oktas), S: wind speed (m/s) at 10 m elevation, and S_o: wind speed of 4.4 m/s, d_f: first experiment day, d_1: last experiment day, D_{ye}: estimated dew yield (mm/day), D_{yc}: dew yield collected experimentally (mm/day), and α: proportionality factor.

Elashmawy and Alshammari [47] proposed and designed a theoretical model for a parabolic concentrator used to force evaporation process from a high concentration desiccant by provided elevated temperature reaches the boiling point. Several sections were done included design of parabolic concentrator, investigated desiccant concentration, and illustrated the used weather data. While the energy analysis was done based on Eqns. 9–12, the parabolic concentrator efficiency and its thermal efficiency are illustrated in Eqns. (13) and (14). After the parabolic concentrator design energy analysis is done, a weather data had been utilized in according with an experimental study.

The energy acquired analysis for the designed model:

$$E_{g,soild} = \sum (mC\Delta T) \quad (9)$$

$$E_{g,water\ liquid} = m_w C_p \Delta T \text{ which is the sensible heat} \quad (10)$$

$$E_{g,water\ vapor} = m_v LH \text{ which is the latent heat} \quad (11)$$

$$E_g = E_{g,soild} + E_{g,water\ liquid} + E_{g,water\ vapor} \quad (12)$$

$$\eta_{PC} = \frac{E_g}{E_R} \text{ the parabolic concentrator efficiency} \quad (13)$$

$$\eta_{th} = \frac{yield\ x\ LH}{E_R} \quad (14)$$

Where $E_{g,soild}$: energy gained by solid tabular solar still components, m is the mass of each tabular solar still solid component, C is the specific heat of each material. $E_{g,water\ liquid}$: energy gained by water at liquid phase, m_w: the liquid water mass, C_p: the specific heat of the water, ΔT: is the temperature difference, $E_{g,water\ vapor}$: energy gained by water at vapor phase, m_v: the vapor water mass, LH: the latent heat of the water, E_g: the total energy gained by all tabular solar still components, η_{PC}: *the efficiency* of parabolic concentrator which is the ratio of the energy gained by tabular solar still to the energy received by parabolic concentrator, η_{th} thermal efficiency which is defined as the ratio of the energy utilized by water to the total solar energy hitting the reflector area.

A vapor compression refrigeration device was designed theoretically and constructed by selecting their component to be suitable for a small family consisting of 4 persons. The theoretical used model is illustrated by Eqns. 15–20, where the P_s and P_a are the water vapor and atmospheric pressures respectively. Moreover, the P_{ssw} *and* P_{ssd} are the saturation pressures at wet bulb and dry bulb temperatures respectively and had determined using Eqns. (18) and (19). According to Eq (20), the collected water ($\dot{m}_w$) is directly depending on the difference of the air moisture between inlet and outlet points through the evaporator $(g_o - g_i)$ [73].

$$g = 0.622 \frac{P_s}{P_a - P_s} \quad (15)$$

$$P_s = P_{ssw} - \frac{1.8\ x\ P_a\ x\ (DBT - WBT)}{2700} \quad (16)$$

$$\varphi = \frac{P_s}{P_{ssd}} \quad (17)$$

$$P_{ssd} = \exp\left(14.4350 - \frac{5333.33}{DBT + 273.14}\right) \quad (18)$$

$$P_{ssw} = \exp\left(14.4350 - \frac{5333.33}{WBT + 273.14}\right) \quad (19)$$

$$\dot{m}_w = \dot{m}_a (g_o - g_i) \quad (20)$$

Where $\dot{m}_a$:air mass flow rate, DBT: Dry bulb temperature, and WBT: Wet bulb temperature.

The following Eqns. 21–23, along with some experiments, were used to design an atmospheric radiation shield and perform heat flux analysis for a steady state [36].

$$\dot{Q}_{dew} = \dot{Q}_{cool} - (\dot{Q}_{sun} + \dot{Q}_{atm}) - \dot{Q}_{conv} \quad (21)$$

$$\text{And } \dot{Q}_{dew} = \dot{m} h_{fg} \quad (22)$$

$$\dot{Q}_{conv} = h_c (T_{amb} - T_{sample}) \quad (23)$$

$\dot{Q}_{atm} = 0$ because the selective emitter is rested on a Styrofoam block.

Where $\dot{Q}_{cool}, \dot{Q}_{sun}, \dot{Q}_{atm}, \dot{Q}_{conv}, \dot{Q}_{dew}$ are radiative cooling, radiative heat gains from the sun, radiative heat gains the atmosphere, Convective heat gains from the air surrounding the emitter, and latent heat gains during dew formation. And $\dot{m}, h_{fg}, T_{amb}, T_{sample}$ are the rate of condensate mass change, the latent heat of condensation, the ambient, emitter temperatures respectively.

The value of the radiative cooling $\dot{Q}_{cool}$ had been studied and changed through maximize the emissivity of the selective emittance, accordingly the radiative heat that acquire from the sun is decreased. To design a water harvesting apparatus based on adsorption materials and to investigate an experimental study, the water collected process has been assumed undergoing the following Eqns. 24–28 [40].

$$q_c = \int_{sunrise}^{sunset} \dot{q}_c(t) dt = A_c \int_{sunrise}^{sunset} \overline{h} (T_a - T_c) dt \quad (24)$$

where t: is the time in seconds, A_c: the condenser surface area, h is the convection heat transfer coefficient in $W/m^2.K$ and T_c is the condenser temperature in Kelvin.

The usage of latent heat to alter the phase is:

$$q_{c,latent} = \omega_{cap}\ q_{st} \quad (25)$$

$$\omega_{cap} = \frac{density\ of\ water\ x\ volume\ of\ water}{mass\ of\ sorbent\ to\ be\ used} \quad (26)$$

Where, ω_{cap} : the mass of the captured water and q_{st}: is the isosteric heat of adsorption which equal 50 kg/mol for the used silica gel and the lowest amount of energy required to release all trapped water after captured it $q_{H,min} > \omega_{cap}\ q_{st}$. Moreover, to determine the needed condenser area and the water yield cycle 's overall efficiency Eqns. (27) and (28) has described that.

$$A_c = \frac{m_{mof}}{\int_{sunrise}^{sunset} \overline{h}(T_a - T_c) dt} \left[(T_a - T_{dew}) (\omega_{cap} C_{p,w}) + \omega_{cap}\ h_{fg} \right] \quad (27)$$

$$\eta_{WHC} = \frac{m_{coll}}{\omega_{rel} m_{Sorb}} = \eta_R . \eta_C \quad (28)$$

Where m_{mof} : *mass of MOF* , $C_{p,w}$ heat capacity of water (J/kg·K) m_{coll}: mass of the collected water (g), m_{Sorb}: masses of the sorbent (kg), ω_{rel}: mass of the released water per unit mass of sorbent (unitless), η_C: water capture efficiency, η_R: water release efficiency

In another research work, desiccant wheels water harvesting units had been presented and simulated. It was made up of several stages of desiccant wheels and ran on renewable energy [46], theoretically the power consumption was assumed to be induced merely by the air fans and had calculated by the developed Eqn. (29). It depends on the volumetric flow rate, G, total pressure drops, $\sum_{i=1}^{m} \Delta P_i$, and the fan efficiency, η_{fan}. Furthermore, the water harvesting rate (WHR) and efficiency (WHE) had been estimated through Eqns. (30) and (31), respectively.

$$P = \sum_{j=1}^{N+1} \frac{G_j \sum_{i=1}^{m} \Delta P_i}{3600\ x\ \eta_{fan}} \quad (29)$$

$$WHR = \frac{G_{pro} \rho (\omega_{pro,2N} - \omega_{pro,exh})}{1000} \quad (30)$$

$$WHE = \frac{WHR}{P} \quad (31)$$

Where $\omega_{pro,2N}$: humidity ratio of processed air after it is humidified by desiccant wheel N, (in g/kg), $\omega_{pro,exh}$: humidity ratio of processed air after it is condensed by the air cooler (in g/kg), and P: energy consumption of the system (in kW).

In an experimental setup, a two-ton refrigeration unit was used to collect atmospheric water content by condensing air. The refrigeration unit was installed in a house with multiple rooms and a corridor, and the air properties were controlled by opening and closing the doors. The collected water was drained into a water tank [74]. Another study proposed a moisture harvesting index (MHI), as shown in Eqn. (32), to evaluate the energy requirements, efficiency, and water delivery rate of water generators that use direct cooling methods for atmospheric water harvesting, in response to the significant energy loss associated with direct cooling methods [75].

$$\mathrm{MHI} = \frac{\mathrm{h_{fg}}}{\mathrm{q^*_{tot}}} = \left(\frac{\mathrm{r_i - r_o}}{\mathrm{h_i - h_o}}\right)\mathrm{h_{fg}} \tag{32}$$

where:

h: is the specific heat enthalpy of the moist air.

q^*_{tot}: the total heat that is needed to remove from air to produce 1 kg of water, which equals the enthalpy difference between its inlet, i, and outlet, o, $(\mathrm{h_i - h_o})$. As well as the enthalpy of condensation h_{fg}.

r: is the air moisture content, in which the difference in air moisture content is $r_i - r_o$.

However, MHI in direct cooling processes always will be less than 1, for instance, when the sensible and latent heat interactions are identical, the MHI will be equal to 0.5, while for warm and humid air the MHI will be high values [75].

Researchers utilized engineering equation solver (EES) software to conduct thermodynamic analysis and predict the amount of harvested water. They introduced actual climate data and solved thermodynamic models to propose and examine a water harvesting apparatus based on an air conditioning cycle combined with a climate chamber. A theoretical model was designed by applying energy and mass balances on each component of the system, Eqns. 33–40, and EES was used to build codes and numerically solve the system of equations. Experimental work was done to simulate the theoretical model using hourly climate data to compare with the results. The study found that increasing the air temperature and relative humidity increased the amount of condensed water [76].

$$\dot{Q}_{evap} = \dot{m}_{air,evap}\left(h_{a,ev,in} - h_{a,ev,out}\right) \tag{33}$$

$$\varepsilon_{evap} = \frac{h_{a,ev,in} - h_{a,ev,out}}{h_{a,ev,in} - h_{a,s}} = \frac{w_{ev,in} - w_{ev,out}}{w_{ev,in} - w_{ev,s}} \tag{34}$$

$$\dot{m}_{water} = \dot{m}_{air,evap}\left(\omega_{ev,inlet} - \omega_{ev,exit}\right) \tag{35}$$

$$\dot{Q}_{evap} = \dot{m}_{ref}(h_1 - h_4) \tag{36}$$

$$\dot{Q}_{cond} = \dot{m}_{ref}(h_2 - h_3) = \dot{m}_{air}C_p\left(T_{cond} - T_{amb,air}\right) \tag{37}$$

$$\dot{W}_{comp} = \dot{m}_{ref}(h_2 - h_1) = \frac{\dot{W}_{comp,s}}{\eta_s} \tag{38}$$

$$\dot{Q}_{cond} - \dot{Q}_{evap} - \dot{W}_{comp} = 0 \tag{39}$$

$$\dot{Q}_{evap} = \dot{m}_{air,evap}\left(h_{a,ev,in} - h_{a,ev,out}\right) \tag{40}$$

Where, $\dot{Q}_{evap}$: evaporator cooling capacity, ε_{evap}: evaporator effectiveness, $h_{a,ev,in}, h_{a,ev,out}$ enthalpy of inlet and outlet air, $w_{ev,in}, w_{ev,out}$ humidity ratio of the inlet and outlet air at evaporating temperatures of refrigerants, $w_{ev,in}, w_{ev,s}$ humidity ratio through evaporator inlet point and surface of evaporator, $\dot{m}_{ref}, \dot{m}_{air,evap}$ refrigerant and air mass flow rates across evaporator respectively. h_1, h_2, h_3, h_4, T_{cond} , η_s, $\dot{Q}_{cond}$ are enthalpies of refrigerants through VCRC, condenser temperature, isentropic efficiency of compressor, rate of heat transfer across condenser, and the isentropic efficiency of compressor.

A study was performed to suggest and test different configurations of air conditioning and humidification-dehumidification system. A mathematical model had been developed and solved analytically using EES under specific assumptions of fixed relative humidity and air temperature values of 50% and 24 °C, respectively [77]. Moreover, the water is condensed at cooling coil temperature. The mass and energy balances equations were applied and carried out through all system component to figure the integrated and needed mathematical relations. Various processes have been schematically plotted in psychrometric charts. Through the analysis of the objectives based on the mathematical model and figured the results, the amount of harvested water increased as the cooling power increased. The governing equations that accommodate the main part of water collected process were accounted by Eqns. 41–45, while the power saving had been calculated using Eqns. 46–49.

$$\mathrm{COP} = \frac{\dot{\mathrm{Q}}_{\mathrm{c,c}}}{\dot{\mathrm{W}}_{\mathrm{c}}} \tag{41}$$

$$\dot{\mathrm{Q}}_{\mathrm{c,c}} = \dot{\mathrm{m}}_{\mathrm{air}}\left(\mathrm{i_{a,i,e} - i_{a,o,e}}\right) \tag{42}$$

$$\dot{\mathrm{m}}_{\mathrm{steam}} = \dot{\mathrm{m}}_{\mathrm{air}}\left(\mathrm{w_{a,o,h} - w_{a,i,h}}\right) \tag{43}$$

$$\dot{\mathrm{m}}_{\mathrm{air}} = \frac{\dot{\mathrm{Q}}_{\mathrm{R}}}{\left(\mathrm{i_{a,R} - i_{a,S}}\right)} \tag{44}$$

$$\dot{\mathrm{m}}_{\mathrm{water}} = \dot{\mathrm{m}}_{\mathrm{air}}\left(\mathrm{w_{a,i,e} - w_{a,o,e}}\right) \tag{45}$$

$$\dot{\mathrm{E}}_{\mathrm{steam,h}} = \dot{\mathrm{m}}_{\mathrm{steam}} * \mathrm{h_{fg}} \tag{46}$$

$$\dot{\mathrm{E}}_{\mathrm{BS}} = \dot{\mathrm{W}}_{\mathrm{c}} + \dot{\mathrm{E}}_{\mathrm{h}} \tag{47}$$

$$\dot{\mathrm{E}}_{\mathrm{PS}} = \dot{\mathrm{W}}_{\mathrm{c}} + \dot{\mathrm{E}}_{\mathrm{pump}} \tag{48}$$

$$\text{Power saving\%} = \frac{\dot{\mathrm{E}}_{\mathrm{BS}} - \dot{\mathrm{E}}_{\mathrm{PS}}}{\dot{\mathrm{E}}_{\mathrm{BS}}} \tag{49}$$

Where COP , $\dot{\mathrm{Q}}_{\mathrm{c,c}}$, $\dot{\mathrm{W}}_{\mathrm{c}}$, $\dot{\mathrm{m}}_{\mathrm{air}}$, $\dot{\mathrm{m}}_{\mathrm{steam}}$, $\mathrm{i_{a,i,e}}$, $\mathrm{i_{a,o,e}}$, $\mathrm{w_{a,o,h}}$, $\mathrm{w_{a,i,h}}$ are coefficient of performance, refrigeration capacity of cooling coils, compressor power, air and steam mass flow rates, air specific enthalpy at inlet and outlet of humidifier respectively. And $\dot{\mathrm{Q}}_{\mathrm{R}}$, $\mathrm{i_{a,R}}, \mathrm{i_{a,S}}, \mathrm{w_{a,i,e}}, \mathrm{w_{a,o,e}}, \dot{\mathrm{E}}_{\mathrm{steam,h}}, \dot{\mathrm{E}}_{\mathrm{BS}}, \dot{\mathrm{E}}_{\mathrm{PS}}$, $\dot{\mathrm{E}}_{\mathrm{Pump}}, \dot{\mathrm{E}}_{\mathrm{h}}$ are refrigeration capacity through room, air specific enthalpy at room and conditioning space supply state, air specific humidity at inlet and outlet points of evaporators, energy of steam, total electrical power of the basic and proposed systems and pump, and humidifier respectively.

A complicated mathematical model was presented and solved analytically using EES. The mathematical model procedure included a design of concentrated photovoltaic thermal unit, sterling engine, absorption cycle, in addition to a cooling and dehumidification system. Two cooling powers were proposed and employed to condense the air water content [78]. The water harvested process had been governed by applying the energy and mass balances equations on the system, however, the main effected parameters were given in the following Eqns. 50 and 51.

$$Q_{evap} = \dot{m}_a\left(h_{air,in} - h_{air,out}\right) - \dot{m}_{water}h_{water} \tag{50}$$

$$\dot{m}_{water} = \dot{m}_{air}(\omega_{in} - \omega_{out}) \tag{51}$$

Where $h_{air,in}, h_{air,out}, h_{water}, \omega_{in}, \omega_{out}$ are air enthalpy at inlet and outlet points, enthalpy of water, humidity ratio at the inlet and outlet points.

4. Advances in water harvesting systems

Advancements in technology have led to innovative solutions in many fields, including the collection of water from the atmosphere. Atmospheric water harvesting systems have gained popularity in recent

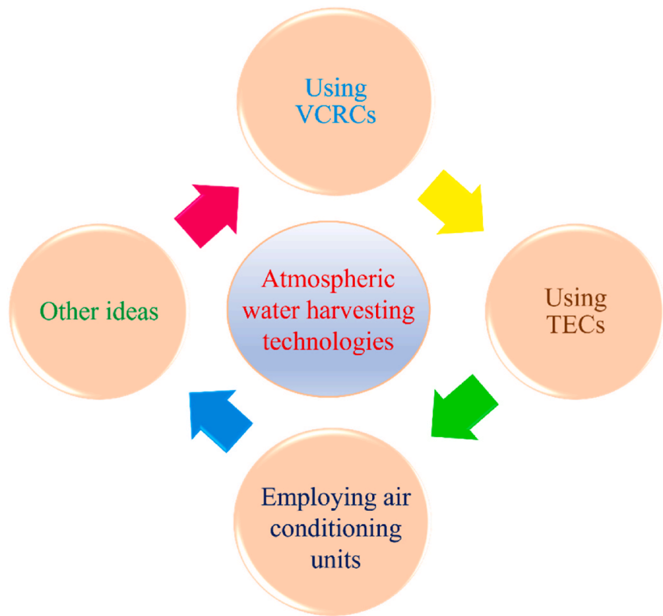

Fig. 8. Approaches and recent advances in water harvesting systems.

years as they offer a sustainable water source in dry regions. This section aims to provide a comprehensive overview of the current state of the art in atmospheric water harvesting systems. The section will be divided into subsections, each focusing on a specific type of atmospheric water harvesting system, summarizing their achievements and features. A conceptual chart (Fig. 8) has been included to aid understanding. This overview provides insights into the various innovations and advancements in atmospheric water harvesting and highlights the challenges and opportunities for future research.

4.1. Atmospheric water harvesting technologies using VCRC

Atmospheric water vapor can be harvested using various techniques, with the most common ones being surface cooling and desiccants [79]. These methods have their advantages and disadvantages, but the focus of current research is on adsorbent-based methods. Despite being widely used, the drawback of surface cooling is that it consumes a significant amount of energy. A vapor compression refrigeration device could harvest 7.9 L/day of water with a relatively low energy consumption. An energy intensity between 220 and 300 Wh/L to produce 22–26 L of water daily is required [73].

In surface cooling, air is cooled to below the dewpoint temperature by a heat pump, causing the water vapor to condense on the heat exchanger surface. Desiccants techniques, on the other hand, work by reducing the water vapor pressure through a desiccant surface, drawing water molecules from atmospheric air [79]. Harvesting atmospheric water requires a substantial amount of energy, approximately 681 kW, since it makes up 4% of the atmosphere's gas mixture by volume and 3% of its mass [80].

The solar-driven hygroscopic water harvesting technique is an old but still effective method of collecting atmospheric water [81]. Most water harvesting devices use vapor compression cycles, but research is focusing on adsorbent-based methods due to their potential energy savings [82]. A theoretical study proposed a solar system that uses concentrated photovoltaic thermal units and Stirling engines to produce electricity and dehumidify water from the air, but it consumes a relatively high amount of energy. A computer model for extracting water from ambient air via a vapor compression cycle was also created. The model was tested in a case study and found to consume between 220 and 300 Wh/L to produce 22–26 L of water daily, The model is shown in Fig. 9 [83].

Employing a VCRC, an atmospheric water gathering device could operate under different weather climate conditions such as humid and moderate, humid and cool, and humid and warm [84]. Table 2 shows the findings after testing the fabricated atmospheric water harvesting device over several climatic conditions.

Table 2
A summary of the main findings in the literature.

Tested Condition	T (°C)	RH (%)	Experimental Water harvested rate (L/h)	Power consumption (kWh/L)
Warm and humid	35	95	1.78	0.75
Moderate and humid	25	75	0.61	2.04
Cool and humid	17	80	1.05	1.22
Warm and dry	40	27	0.28	4.71
Moderate and dry	22	45	0.30	4.40
Moderate	22	50	0.35	4.11
Warm and moderate humid	35	70	1.35	1.03

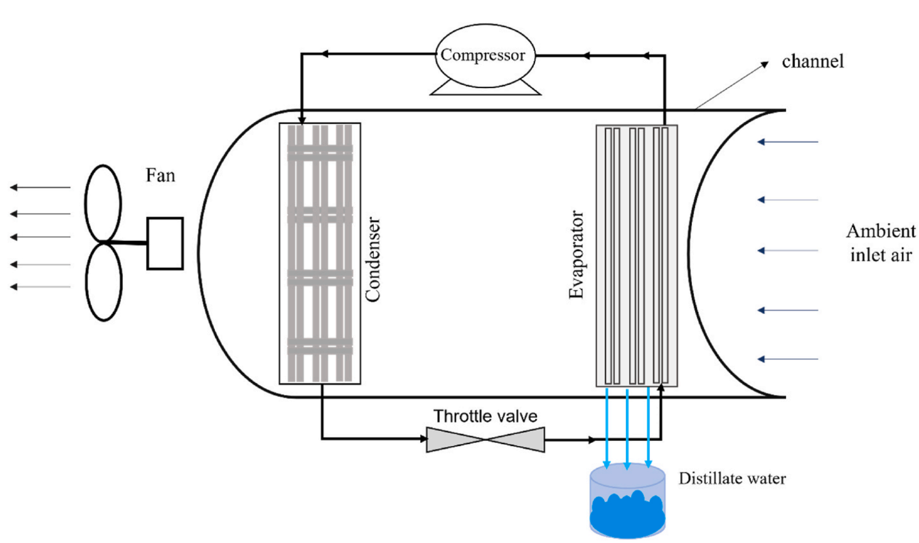

Fig. 9. Schematic of the developed model of the vapor compression cycle.

Table 3
A recap of recent research focus in the area of VCRC for gathering atmospheric water.

Ref.	Remarks
[85]	An experimental study with focusing on the number of evaporator rows in harvesting atmospheric water.
[73]	A water device to meet drinking water requirements of 4 people and provided 7.9 L/day.
[78]	Theoretical model relying on thermal photovoltaic panels with Stirling engine to run VCRC and supply 26.28 L/h of water as a maximum output.
[83]	A computer model of VCRC was built and evaluated; it was able to get 22–26 L/day.
[84]	Different weather conditions were simulated and tested by VCRC with a climate chamber, the highest water was harvested under the warm and humid environment which was 1.78 L/h.

According to Table 2, the warm and humid conditions had the lowest power usage of 0.75 kW/L, while the warm and dry conditions had the highest value of 4.71 kW/L. It can be highlighted that hot and humid districts are the greatest and best conditions for harvesting atmospheric water [84].

The VCRC has been identified as one of the most effective methods for harvesting atmospheric water when a large volume of water is required. Additionally, the VCRC is highly adaptable and can be designed to meet specific requirements, such as the desired collection rate or the environmental conditions. A quick review of the most current literature that has recently looked at the VCRC method is provided in Table 3, which summarizes the key findings of various studies that have investigated the VCRCs effectiveness in harvesting atmospheric water. These studies have shown that the VCRC is capable of collecting large volumes of water under certain conditions, and with further research and development, it has the potential to become a major contributor to solving the world's water scarcity problem.

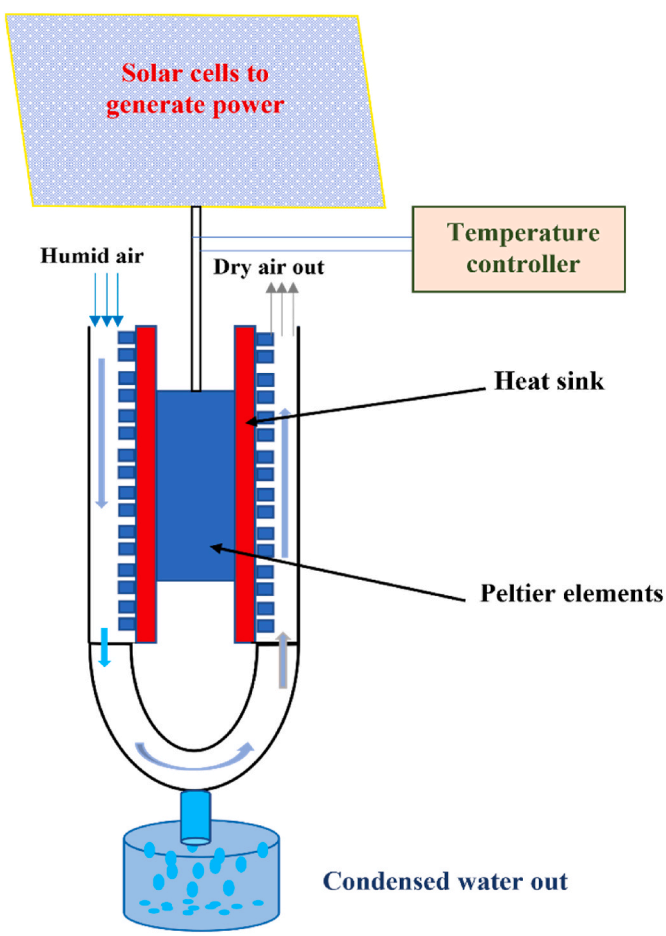

Fig. 10. Schematic of the developed clean water harvester.

4.2. Employing solar energy and TECs as a water harvesting method

TECs or Peltier devices are compact and efficient cooling and heating devices that use the Peltier effect to produce cooling by passing an electric current between two sides, resulting in the dissipation of heat generated at the hot end. These devices are frequently used for cooling, and their energy source can be made sustainable and clean through the use of solar panels and batteries. The semi-conductive materials used in TECs have a temperature range of −130 °C–90 °C, making them ideal for harvesting atmospheric water [86]. Several studies have explored the use of TECs in capturing atmospheric water, with one study presenting a TEC-based device powered by a solar cell that was designed to be used for irrigation and in isolated regions. A schematic of clean water harvester is presented in Fig. 10 [87]. Computational fluid dynamics was used to investigate the potential of using a TEC powered by solar panels for harvesting atmospheric water, with results demonstrating the Peltier effect's potential for capturing atmospheric water [88].

Various methods of using TECs or Peltier devices for harvesting atmospheric water with solar energy have been investigated [89]. By analyzing parameters such as temperature, relative humidity, and current, water production efficiency was evaluated, and electrical systems were designed to collect water from moist air and dissipate foggy conditions [90]. Agricultural applications for atmospheric water generators have also been explored, with one design collecting one gallon of water per week in an environment with 60% relative humidity and 29.5 °C [91]. A water condenser unit based on the thermoelectric effect was introduced for providing water for young trees during drought periods in arid climates [92]. A TEC system integrated with a solar distiller was also developed and tested in Beirut's humid climate, providing 10 L or more of fresh water per day [93]. A schematic of the atmospheric water harvesting system that is based on utilizing TEC is shown in Fig. 11.

To produce portable freshwater in coastal regions with relative humidity over 60%, a water generator with ten thermoelectric modules was designed. The modules contained an internal heat sink on the cold side, and the experiments revealed an 81% increase in water production compared to modules without a heat sink [94]. The use of thermoelectric coolers has been extensively researched as an atmospheric water generator, and various designs have been found to be effective. The Peltier effect is promising for efficient water production, with optimal conditions at high relative humidity levels 80–84% and when powered by renewable energy sources such as solar panels. Further research and development is necessary to improve efficiency and scale of production [95]. An experimental and numerical study was performed utilizing a $4x4\ cm^2$ one thermoelectric module to design and evaluate a small prototype for water production. The prototype had been positioned inside a duct, and a condensing surface with has cone shape had been designed used to make the condensed droplets fall easily. The results revealed that Increasing the air flow rate at the heat sink side had led to increasing water production [96].

Water harvesting devices based on TEC have been used in a variety of prototypes and are ideal for situations where low cost and small amounts of water are required. Recent research has focused on combining TEC with solar energy to maximize water collection from the air. Table 4 provides a brief summary of the amount of water collected in the most recent TEC-based studies. TEC-based water harvesting devices work by using the temperature difference between two materials to generate an electric current, which then powers a cooling element. This cooling element can lower the temperature of a surface below the dew point, causing moisture in the air to condense on the surface and be collected. The efficiency of TEC-based water harvesting devices depends on several factors, including the size of the cooling element, the surface area of the

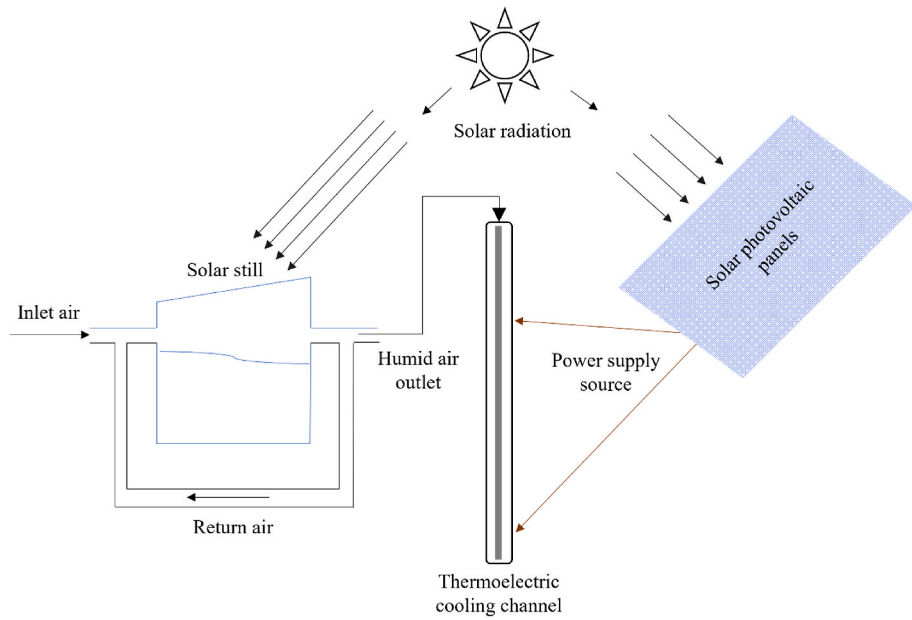

Fig. 11. The solar distiller-TEC water harvesting unit.

Table 4
Research summaries findings of some recent studies based on utilizing TEC modules.

Ref.	Method	Capacity	RH
[88]	Simulation using CFD	3.16 $g_{water}/m^3 of\ air$ and was calculated using the psychometric chart 0.32 L/h was obtained theoretically	NA
[89]	Simulation	26 ml/h	75%
[91]	Experiment	0.21 L/day	69.6%
[92]	Experiment	2–4 ml/h	NA
[97]	Experiment	25.1 g/h 11.2 g/h	92.7% 67.8%
[93]	Simulation	14.09 L/day in August 10.06 L/day in October	NA
[94]	Experiment	240 ml/day for a total of 10 h	90%
[98]	Simulation	66 ml/h	80%
[95]	Experimental	19 $L/m^2.day$	80–85%
[96]	Experimental	20 ml/h	75%

condensing surface, and the temperature difference between the cooling element and the surrounding air. While TEC-based water harvesting devices have shown promise in laboratory settings, further research is needed to optimize their performance and to develop practical applications for real-world use.

4.3. Air conditioning unit water harvesting

The field of atmospheric water harvesting for air conditioning and energy efficiency is still being researched, with the main considerations being the device's water production rate and energy consumption. The amount of water that could be collected from a two-ton air conditioning system was predicted and it was found that the amount collected was related to inside and outside room humidity values, with an average of 2.07 L/h [99]. Different configurations of air conditioning and humidification-dehumidification system was suggested and tested. The mass and energy balance equations were applied and carried out through all system components to figure out the integrated and needed mathematical relations. Various processes have been schematically plotted in a psychrometric chart. The amount of harvested water had been increased as the cooling power increased [77]. A 1.5-ton split air conditioning system, shown in Fig. 12, could be used as a supplementary source of water [76]. An integrated system aimed at providing cooled air and water for a hotel and operating on a reverse compression cycle and extracting output heat flow for domestic hot water through a plate fins heat exchanger, as shown in Fig. 13 [100].

The potential for atmospheric water harvesting to improve energy efficiency and provide a supplementary source of water has been investigated. One study analyzed an Heating, Ventilation and Air-Conditioning (HVAC) system for a hotel in Abu Dhabi that utilized condensed water for domestic use, collecting an average of 10.2 m^3/day of atmospheric water [101]. Another study compared a typical HVAC system to an integrated one that covered 56.4% of a hotel's water needs and resulted in a 19% cost reduction for water [102]. An advanced system for producing freshwater in hot and humid climates was also designed and investigated, which produced 250 kg/h of freshwater with an airflow rate of 3 kg/s [103].

As previously stated, some air conditioning units are capable of generating sufficient water for specific purposes. This presents numerous possibilities for utilizing the collected water effectively in buildings and homes where air conditioning is regularly used. Table 5 provides a brief overview of recent developments in this area. The current research highlights the importance of water production rate and energy consumption in atmospheric water harvesting for air cooling and energy efficiency. However, further research is necessary to maximize the amount of water harvested at the lowest possible cost.

4.4. An integrated and advanced techniques for harvesting atmospheric water

Research models focused on atmospheric water collection techniques have been developed to reduce energy consumption and increase the efficiency of water harvesting. One such method is the electrostatic water nucleation technique which uses electrospray to improve the condensation process by increasing the movement of water molecules to the cooling surface [104]. Another innovative technology that has been developed uses electrostatic boosting of phase change variation operations during condensation to harvest atmospheric moisture. The prototype consists of a moisture separator and a condenser and can produce 7.5 ml/h of water [105]. A new integrated water-harvested device based on a silica gel desiccant wheel with a vapor compression cycle has been tested in arid climates and showed efficient water harvesting [106]. Another integrated system has been designed to produce water from the

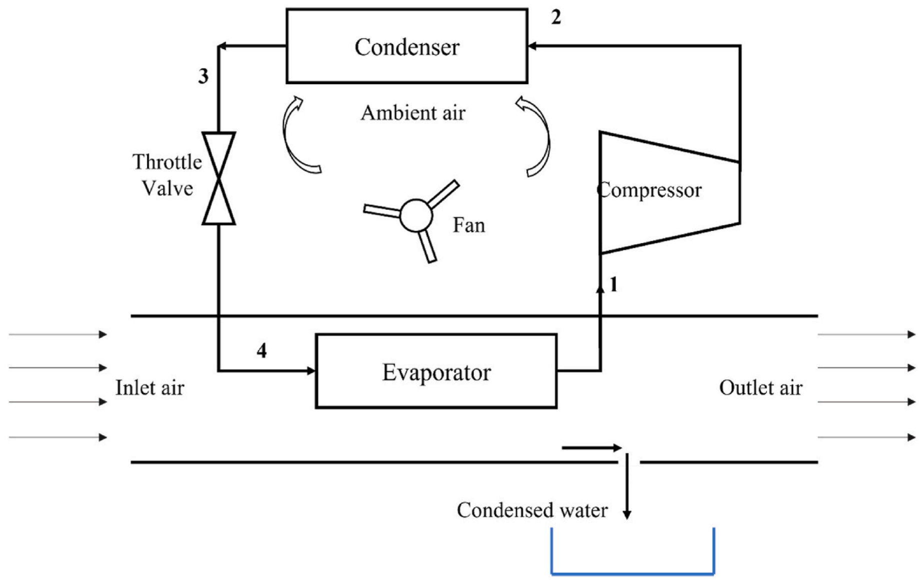

Fig. 12. Schematic of a water harvesting device based on an air condition unit.

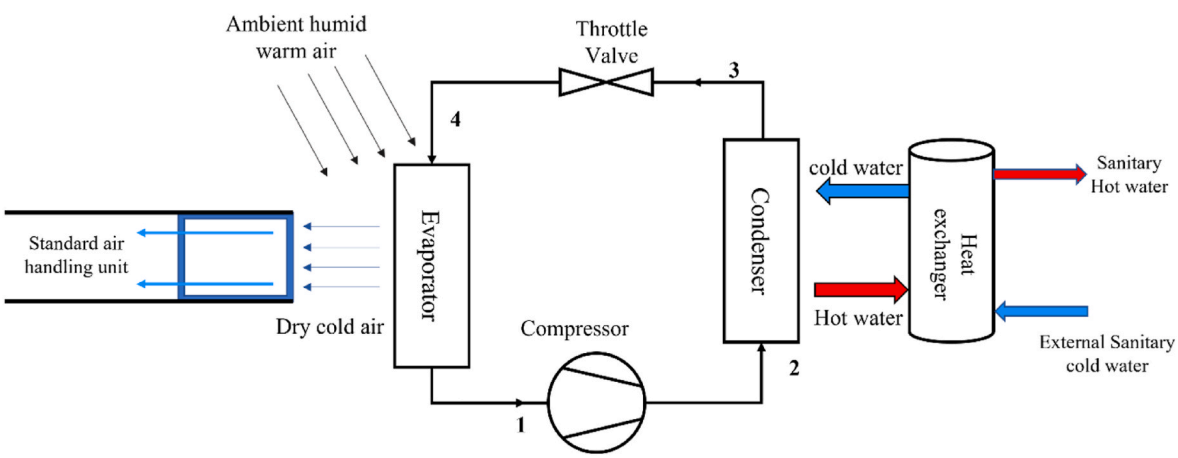

Fig. 13. Schematic of the reverse compression cycle.

Table 5
Remarks regarding the use of an air conditioner to collect atmospheric water.

Ref.	System used	The studied area	RH	Productivity
[102]	Integrated HVAC system consisting of modules, for each one the airflow rate was 30000 m^3/h at cooling coils	Case study in Abu Dhabi	NA	Maximum capacity in September was 425 kg/h
[77]	Steam humidifier for increased air humidity and cooling coils for dehumidification	For hot and dry areas	NA	375 kg/h
[103]	Humidification by spray tower then dehumidification to condense air humidity	For hot and humid zones	NA	100 kg/h when the air flow rate is 1 kg/s, and 300 kg/h for 3 kg/s air flow rate
[76]	1.5-ton split air conditioner unit	Dharan, Saudi Arabia	15–90%	70.1 kg/day in August
[99]	2-ton air conditioner unit	Kingdom of Bahrain	62.01%	2.49 L/h in September

atmosphere using ammonia absorption refrigeration cycle, saline water desalination cycle, and air dehumidification cycle, which showed valuable performance and was not greatly influenced by changes in air pressure [107]. In attempt to give more realistic water harvester prototypes, an ammonia vapor absorption system was constructed and tested experimentally in Jordan. The airflow rate and ambient temperature went through analysis, throughout the designed system can produce 10 L/day of water with a specific energy consumption of 9 kWh/L. It is a promising technology that can be utilized with renewable energy to produce water, but it needs more development to lower power use [108]. An integrated system based on desiccant wheels with evaporator coolers, as shown in Fig. 14, has also been presented to supply cooled air and condensed water [109].

These various research models have been developed to improve the process of atmospheric water collection. Techniques such as electrostatic water nucleation and the use of electrospray have been introduced to reduce energy consumption and increase efficiency in water harvesting. These advances in technology are helping to improve the process of atmospheric water collection and making it more efficient and effective.

Research has been conducted on atmospheric water harvesting systems to improve access to water in dry areas [110]. These systems

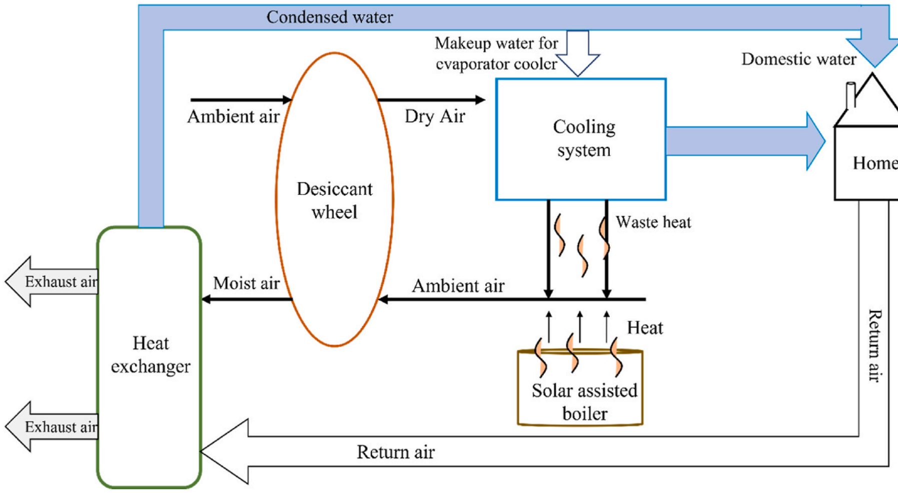

Fig. 14. Integrated desiccant wheel and cooling system.

include using return air to cool and condense water [111], condenser chambers with Peltier modules, and solar-powered plants [112]. Studies have also explored modifications to improve water harvesting efficiency, such as using cascading structures, coating with MIL-101(Cr) [113], and desiccant materials in solar still prototypes, for example, in Kafrelsheikh city, Egypt, an experimental for a modified solar still prototype had been constructed and investigated. The modification had been described and represented by adding four longitudinal fins and gravel inside the prototype. Several fins had been mounted inside the solar still to increase surface area, furthermore a layer of silica gel had been utilized as absorbent material. An observable improvement has been shown in the system efficiency and the collected water had reached 400 ml/m^2 [114]. On the other hand, in a nontraditional study, a humidification process using low water quality to obtain humidified air, then evaporator coils which are represented the dehumidification process were used to collect some atmospheric water. A window air conditioner was used for the experiment. The amount of water collected was discovered to be dependent on the volume inflow rate, cooling coil capacity, moisture, and the time the air remains relatively inside the air conditioning unit. Furthermore, the harvested water was of high quality in comparison to the poor water quality used to humidify the air, and the average collected water volume was 0.3 ml⁄s [115]. Based on an experiment carried out in Doha, Qatar, a self-sustaining energy system has been developed to produce fresh water. One household's water need can be met by a dehumidifier system that is 1 m^3 in size. The volume of freshwater obtained each day can reach 15 L in situations with a relative humidity of 50–70% [116]. Another study was conducted to compare traditional water harvesting cooling systems to a system that uses a desiccant wheel to humidify the air [117].

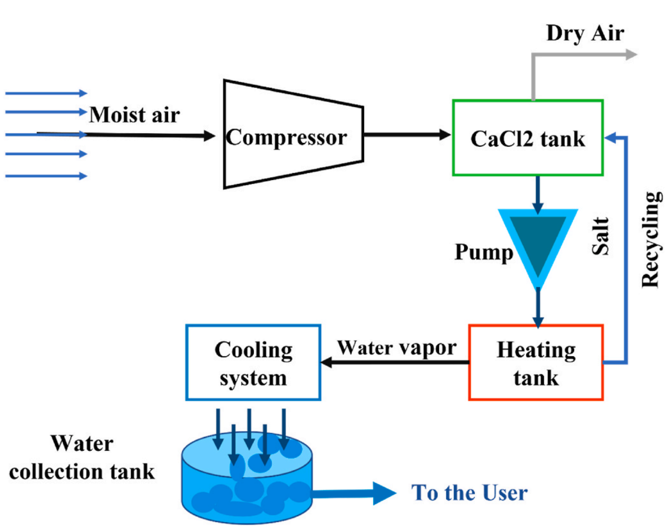

Fig. 15. Water harvesting generator by using $CaCL_2$.

An integrated water harvesting system that relies on the use of $CaCL_2$ salts to absorb air moisture and a cooling system to collect the distilled water has been proposed, as shown in Fig. 15. The objective of this device is to provide drinkable water through an efficient and cost-effective means. This method involves the absorption of water vapor by $CaCL_2$ salts, which are then heated to release the water molecules, producing a high concentration of moisture in the air. This moist air is then cooled using a cooling system, which condenses the water vapor and collects it in a reservoir for use. This integrated water harvesting system has been shown to be effective in producing significant quantities of distilled water and has the potential to be used in regions with low humidity, where traditional atmospheric water harvesting methods are less effective [118]. Nonetheless, further research is needed to optimize the performance of this system and to reduce its energy consumption, which is a critical factor for its widespread adoption.

4.5. Renewable energy sources in water harvesting

Renewable energy can be classified into several types based on their source and conversion process, including solar, wind, hydro, geothermal, and biomass energy. Each type of renewable energy has its own unique advantages and limitations. Solar energy is derived from the sun's radiation and can be converted into electrical energy using photovoltaic cells. It is a widely available and abundant source of energy, but its efficiency is dependent on factors such as climate, time of day, and location.

Wind energy is generated by the movement of air and can be converted into electrical energy using turbines. It is a relatively mature technology, with many large-scale wind farms currently in operation around the world. However, wind energy is also dependent on location and weather conditions. Hydro energy is produced by the force of falling water and can be harnessed by turbines. It is a highly efficient and reliable source of energy, but its availability is limited to areas with sufficient water resources. Geothermal energy is derived from the heat of the earth's core and can be used for both heating and electrical generation. It is a relatively clean and sustainable source of energy, but its availability is limited to areas with geothermal activity. Biomass energy is generated from organic materials such as wood, crops, and animal waste. It can be converted into electricity or used as a fuel source for heating and transportation. However, it can also have negative environmental impacts such as deforestation and greenhouse gas emissions.

When considering the choice of renewable energy solutions, it is important to weigh the advantages and limitations of each type, as well as factors such as cost, availability, and environmental impact. A systematic approach can be used to evaluate and compare different renewable energy solutions based on these factors, ultimately leading to the selection of the most appropriate solution for a given situation.

4.5.1. Wind- and solar powered water harvesting systems

Recent research has explored novel methods of utilizing wind turbine generators to produce both cooling and freshwater. One approach involves had been exploiting dissipated heat from wind turbines for running an absorption refrigeration cycle that uses different water-nanoparticle mixtures for heat transfer, among the three mixtures that were used to produce a high cooling effect and freshwater, the copper/water mixture had the best performance [119]. Another company is proposing a wind turbine system that can produce both electricity and drinkable water for remote areas using a vapor compression refrigeration cycle [120]. Additionally, a self-atmospheric water harvesting condenser powered by a wind turbine has been proposed and tested, as well as a solar chimney, as shown in Fig. 16, for energy and water harvesting that uses black pipes to heat the air and water. Results from these studies demonstrate the potential for utilizing wind turbines to address water scarcity in remote areas [121].

The use of spiral wind turbine and Internet of thing (IoT) technology in water harvesting has shown promising results. The wind and solar energy provide the necessary power for the system to function, and the copper condenser is used to condense the humid air. If the wind speed is insufficient, the system can use direct current (DC) motors to drive the turbine. The experimental results indicate that the system is capable of harvesting between 3.39 and 11.2 L of water per day. The design concept is illustrated in Fig. 17 and demonstrates a sustainable solution for water scarcity in remote areas [122,123].

Looking at the issue from a different perspective, wind energy harvesting systems offer a considerable amount of power. Several studies have concentrated on utilizing the excess energy generated by wind turbines to operate active water harvesting devices. Moreover, some researchers have explored the possibility of integrating solar chimneys with wind turbines to collect and condense atmospheric water.

4.6. Biomass-powered atmospheric water harvesting

The use of different energy sources, such as flared gases, natural oil and gas, and biomass, to produce power for water harvesting systems had been studied. For example, flared gas from Bakken Shales and Eagle

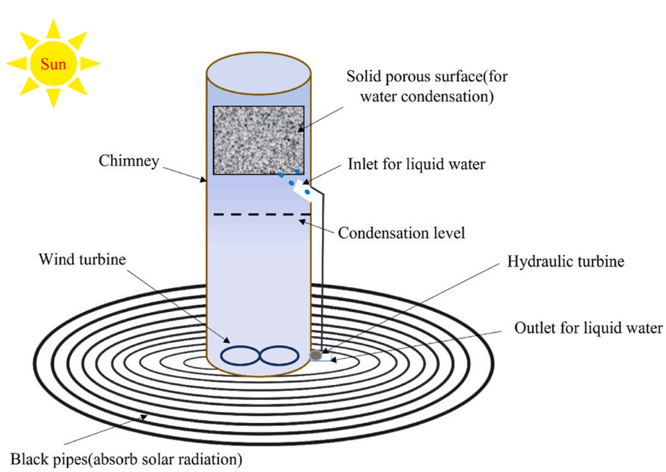

Fig. 16. Solar chimney and wind turbine for water harvesting.

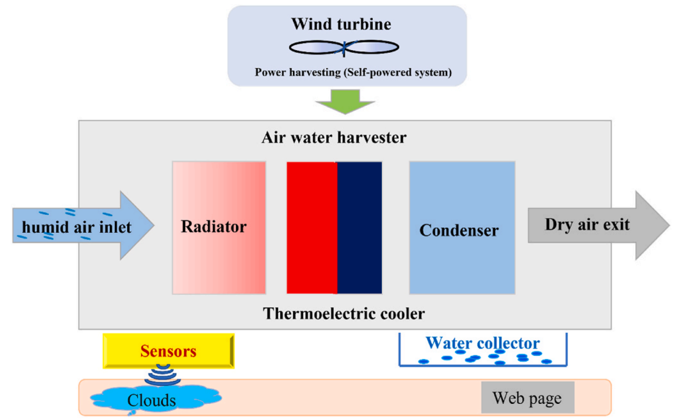

Fig. 17. Smart water harvesting system.

Ford can meet 60% and 15% of yearly water needs, respectively [124]. Natural gas from oil fields and landfills can be used to power refrigeration cycles that condense moisture from the atmosphere, reducing the cost of water production and mitigating the greenhouse effect caused by gas release [125], some researchers found a way to use the remaining and unused biomass for atmospheric water harvesting by burning the biomass and running a harvesting system that operates on a refrigeration system. The work was carried out in an area in India and the system can provide 8000-1200 L of water by using 1000 kg of biomass [126]. Using flared gas to run a refrigeration cycle compressor to condense atmospheric water, instead of wasting it, was proposed and an adsorption refrigeration cycle powered by a gasifier using crop remnants was studied, as shown in Fig. 18, and demonstrated acceptable efficiency, meeting the water needs of 19% of Sri Lanka's population, which amounts to nearly 10 billion liters of water per year [127,128].

In conclusion, biomass is a potential source of large amounts of energy that is often wasted. Some methods have been developed to harness this energy by burning biomass, flared gases, and other forms of lost energy to power VCRC-based water harvesting devices.

4.7. Environmentally and innovatively inspired technologies for harvesting atmospheric water

The development of innovative water harvesting systems that utilize atmospheric air and natural energy sources has been a focus of recent studies. One such system designed for green roofs [129] relies on a diaphanous mesh and a simple dew collector powered by solar energy to gather atmospheric water. Another study [130] discovered that the shape and nature of wheat plant leaf play a crucial role in capturing fog and air moisture, with inwardly rolled surfaces being the most effective. Water harvesting towers [131], which capture water vapor and rely on solar energy for their operation, have also been proposed as a potential solution for providing water supply in disaster and war zones. Additionally, a copper and ethanolamine composite were fabricated and tested for atmospheric water harvesting, with a fully automated system, Smart Farm, as shown in Fig. 19, being developed using this composite material and relying solely on solar energy for its operation [132].

Smart Farm and the desert water harvesting system are innovative solutions to address the global challenge of food insecurity and water scarcity. By utilizing solar energy, these systems are not only sustainable but also environmentally friendly. The Smart Farm system is designed for urban agriculture, reducing carbon footprint and dependence on traditional freshwater resources [132]. The desert water harvesting system can be used in arid regions to efficiently harvest soil water, providing a much-needed source of water for agriculture [133]. Both systems have the potential to be scaled up and play a significant role in ensuring food security and water access for communities around the world. A schematic of soil water harvesting is presented in Fig. 20.

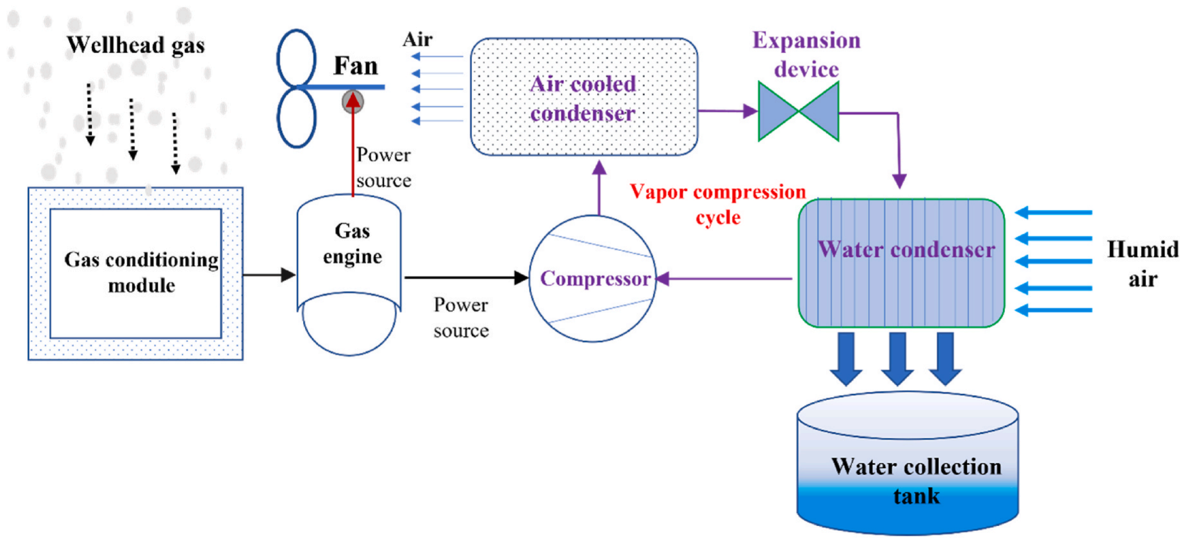

Fig. 18. A gas engine-based atmospheric water harvesting.

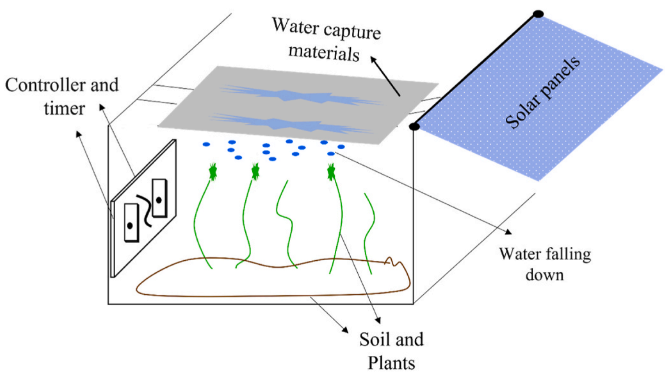

Fig. 19. Schematic of the suggested SmartFarm.

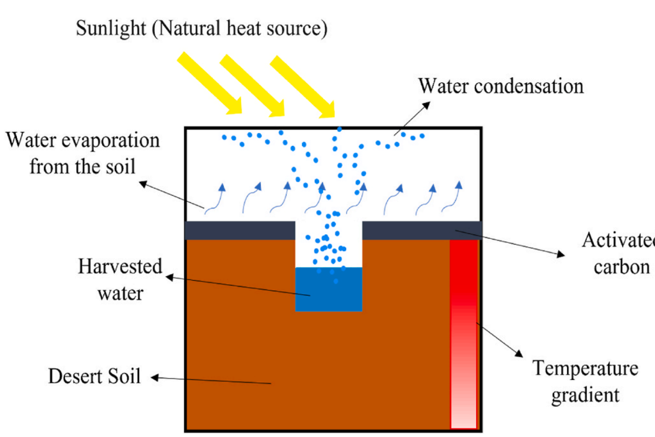

Fig. 20. Soil water harvesting system.

Towards the conclusion of this section, attention was given to the application of atmospheric water in agriculture. One study investigated the most effective plant leaf for collecting dew water, while others proposed the concept of smart farms that function as self-irrigating systems by harnessing moisture from the air. Additionally, other research has explored the possibility of harvesting water from soil through heating and evaporation methods.

4.8. Hybrid fuel cell and vapor compression cycle system for water harvesting and power generation

A hybrid system designed for water harvesting and power generation was developed and investigated, comprising a fuel cell and a vapor compression cycle that uses the fuel cell's exhaust gas for water production. The simulation model showed a 50% improvement in water gathered using a 2-kW fuel cell at 75% relative humidity [134]. The system reduces reliance on external power sources by utilizing the wasted electrochemical heat and water from the fuel cell, as shown in Fig. 21. With no external solar heating source, the 2-kW fuel cell was able to harvest 1800 g/h of water when the airflow rate was 12.5 g, presenting a promising alternative for simultaneous water harvesting and power generation [135].

A multi-function desalination system has been proposed that generates electrical power, potable water, and cold energy. The system consists of a hydrolysis reactor, fuel cell, evaporator, condenser, heat exchanger, water recovery device, photovoltaic panel, and TEC. It provides two sources of water: seawater desalinated through the hydrolysis reactor with the fuel cell and water harvested using a heat exchanger, TEC, water recovery device, and photovoltaic panels. The results show a 15.5% increase in humidity in wet air and 1.8 times increase in condensed water compared to using ambient air. The highest water production rate achieved was 11.10 kg/h with a power consumption of 880 Wh/kg [136]. A hydrogen fuel cell prototype was studied to provide clean drinking water to a typical household, with test results showing that water collected from the fuel cell can be used for human consumption [137]. The produced water by the fuel cell can be expressed as a reference (Buie et al., 2006, [138]):

$$Q_{water} = \frac{j\ X\ A_{FC}\ X\ M_w}{2\ X\ F\ X\ \rho_w} \tag{52}$$

Where j: fuel cell current density, A_{FC}: the fuel cell area, F: Faraday's constant, ρ_w and M_w are the density and molecular weight of liquid water, respectively.

To highlight the importance of harvesting atmospheric water for human consumption, various efforts are being made. With fuel cells commonly producing heat and exhaust gases, there have been attempts to extract drinking water from the atmosphere. Research has shown that fuel cells can indeed be utilized for water harvesting, although more research may be necessary to improve the amount of water produced.

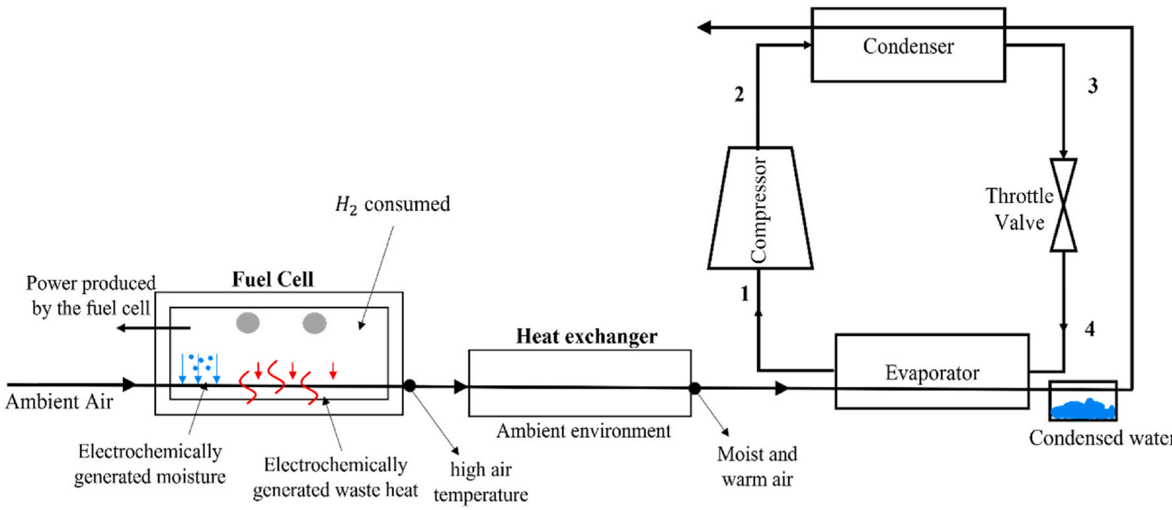

Fig. 21. Using fuel cells for enhancing water production via the condensation process.

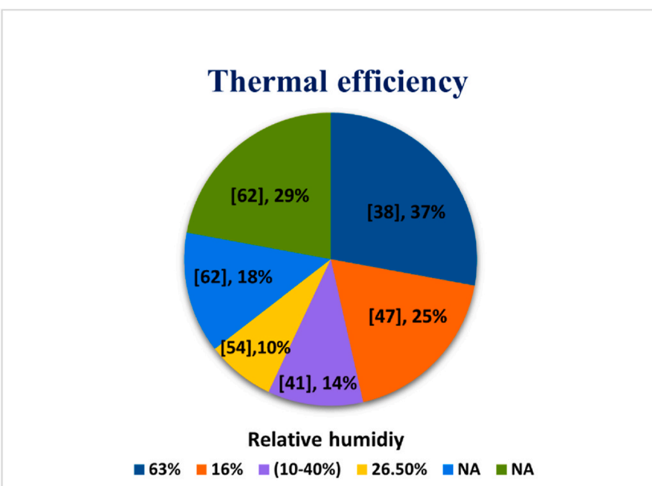

Fig. 22. Thermal efficiency vs the harvested relative humidity.

5. Progress in the assessment of atmospheric water harvesting systems

Water harvesting is a process of collecting atmospheric water through condensation, using power sources such as vapor compression cycles or thermoelectric coolers. While the efficiency and potability of harvested water remain concerns, the process can still benefit water-scarce areas for non-drinking purposes like irrigation. However, current sorption water harvester prototypes have low water yields and may not meet household water demand. Fig. 22 depicts the thermal efficiencies of some sorption-based water harvesting systems.

In recent studies, the performance of water harvesting systems has been analyzed using two metrics: water harvesting rate (WHR) and unit power consumption (UPC). The WHR represents the amount of water harvested per hour and is calculated by dividing the harvested water mass by the operating hours. Eqns. (53) and (54) show how much water is harvested per hour and how much power is consumed per unit mass of water.

$$\text{Water harvesting rate (WHR)} = \frac{m_w}{h_r} \text{ in } kg/h \tag{53}$$

$$\text{Unit power consumption (UPC)} = \frac{P}{m_w} \text{ in } kWh/kg \tag{54}$$

Where:

m_w: harvested water mass, kg, h_r: operating hours, hour, and P: is the consumed power, kW.

The UPC represents the amount of power consumed per unit mass of water. A high WHR and low UPC is ideal for any water harvesting system as it would mean that a large amount of water is harvested while using minimal power [86]. The potential for atmospheric water harvesting using solar-driven devices was evaluated using Google Earth Engine and global data. The results showed that a device with a 1-m square surface area could produce 0.2–2.5 L of water per kilowatt-hour under relative humidity conditions ranging from 30 to 90% [139]. A case study in Matehuala, Mexico showed that an atmospheric water generator could help address the issue of water scarcity and pollution in the region. The study found that the daily yield of water ranged from 3.9 to 18 L, at a cost of 0.038 USD per liter [140].

An experimental study was conducted to assess the average amount of water harvested and electrical energy input to atmospheric water harvesting systems. The experiments were conducted under different climate conditions, including warm and humid, moderate, and cold and dry. The results showed that the average amount of water produced per unit of energy consumption was 1.02 kWh/L for warm humid climates and 6.23 kWh/L for cold humid climates [141]. In Fig. 23, the water harvest rate vs air flow rate of previously published works on atmospheric water harvesting systems is presented.

A study in Chile [142] found that dew water harvesting could collect an average of 1.9 L per day on 56.1% of recorded days. Meanwhile, in Kenya, fog and air humidity harvesting was found to be a low-cost solution for providing potable water and improving visibility in certain regions [143]. Another study had been conducted to assess the potential of harvesting atmospheric water in Thailand using the direct cooling method, the meteorological data had been used in analysis and the results are shown that about 0.97–1.30 L/h of water can harvest [144]. However, challenges in terms of efficiency, cost, and water yield remain [145]. A small scale air-cooled desiccant wheel dehumidifier based on silica gel was developed and evaluated through simulation, producing the highest water yield in Abu Dhabi [146]. A comparative analysis of compressor and desiccant systems showed that the compressor system was more efficient in producing water [147]. Finally, a review of atmospheric water harvesting methods concluded that the sorption/desorption method was highly recommended for tropical buildings, while the radiation dew condensation method was limited in its water production [148].

Another comparison table for the water capacity of the main water harvesting methods is shown in Table 6, which has been carefully

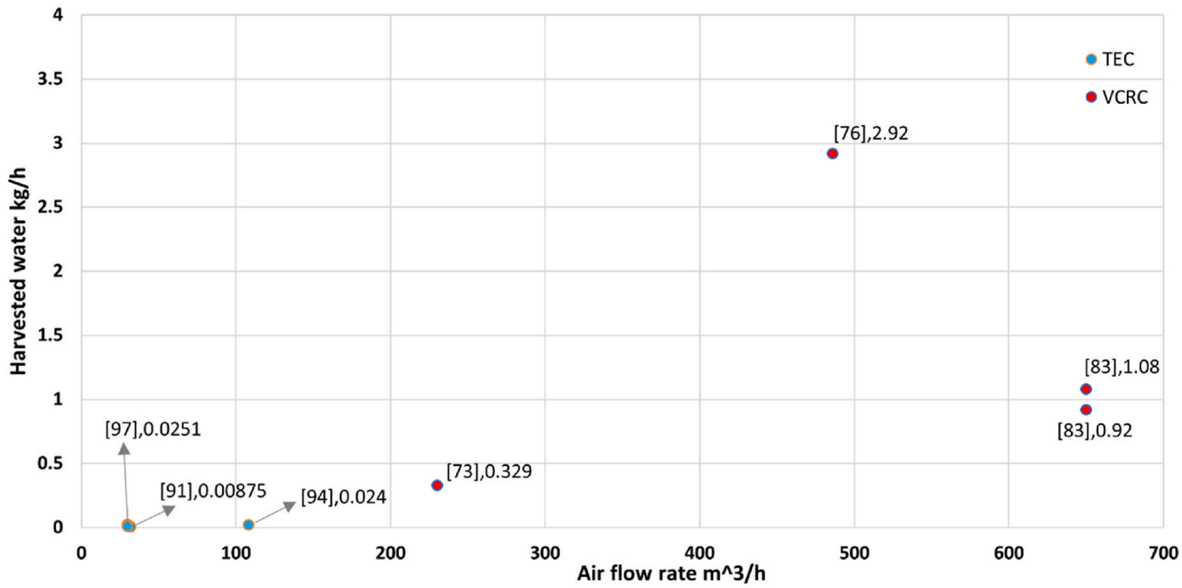

Fig. 23. Harvested water vs air flow rate.

Table 6
Water capacity for a comprehensive sample of references.

Method	Ref.	Amount of water per day (L/day)	Method	Ref.	Amount of water per day (L/day)
Sorption	[54]	1.06	TEC	[91]	0.21
Sorption	[58]	1.49	TEC	[93]	14.09
Sorption	[51]	2.89	TEC	[93]	10.06
Sorption	[60]	2.32	TEC	[98]	1.584
Sorption	[60]	1.235	TEC	[89]	0.624
Sorption	[61]	1.01	TEC	[97]	0.6024
VCRC	[73]	7.9	TEC	[97]	0.2688
VCRC	[83]	26			
VCRC	[84]	42.72			

illustrated and plotted by reviewing the references and ensuring consistency. This table provides an excellent comparison in terms of the daily amount of atmospheric water that can be captured. Based on the data, it is concluded that active water harvesting techniques, and those based on VCRC, have the greatest capacity for capturing air moisture.

In scientific terms, the variation in the amount of harvested water can be attributed to the following factors. VCRC and TEC-based atmospheric water harvesting systems can handle large air flow rates in a short time. These devices rely on cooling the air to the dew point temperature, and once the air comes in contact with a cold surface, water generation starts immediately. On the other hand, using sorbent materials to harvest atmospheric water is a different approach and largely depends on the capabilities of the sorbent materials. Compared to the aforementioned devices, the amount of air moisture captured by the sorbents is relatively small, and it takes more time to capture and regenerate the water by desorbing the water molecules.

6. Conclusions

Atmospheric air harvesting is a method of collecting water from the atmosphere by condensation, which requires an electrical source. The performance of these systems is measured by the WHR and the UPC. Ideally, the systems should have a high WHR and low UPC. The feasibility of atmospheric water harvesting has been studied in different regions and climates, such as Mexico, Chile, Kenya, and Saudi Arabia. These studies have shown that the amount of harvested water and energy consumption vary based on the climate and other factors such as airspeed. In addition, different systems have been developed for water harvesting, including evaporator coils, desiccant wheel dehumidifiers, and others. The challenges associated with atmospheric water harvesting include improving the yield and ensuring water potability, as well as addressing the cost and efficiency of the systems. However, this technology has the potential to provide safely managed drinking water for billions of people worldwide and can also be used for other purposes such as farm irrigation and reducing pressure on traditional water sources.

In conclusion, this work on atmospheric water harvesting presents a comprehensive overview of the various harvesting techniques, their performance, and feasibility. The findings show that different harvesting systems have their own advantages and disadvantages, and the choice of system depends on the specific environmental conditions, such as temperature and relative humidity. The mathematical and thermodynamic models presented in the review can help in estimating the water yield and assist in the selection of the best harvesting technology. The comparative analysis of the amount of water harvested by different technologies provides valuable insights into the performance of the different systems. Overall, the review highlights the importance of continued research in the field of atmospheric water harvesting, particularly in developing new and improved technologies, and the optimization of existing systems to increase their water harvesting potential.

7. Recommendations

Water scarcity is a widespread global issue, so alternatives had to be devised. As a result, given that the atmosphere is a huge available source of water, this study was presented on the topic of atmospheric water harvesting. Water harvesting systems of various types were mentioned and explained, including the use of air moisture-absorbent materials, VCRC, TEC, and many other systems. We recommend the following points:

- Concerning the dew harvester, some research has been conducted on the properties of surfaces on which dew condenses and the possibility of improving them. It is advised in this context to investigate the possibility of using solar cells to harvest dew water and energy simultaneously.
- In terms of fog harvesting, and because it is limited to specific regions, it is recommended to design networks capable of capturing and rapidly releasing water particles into water collectors in order to

obtain the greatest amount of water in the shortest amount of time and to achieve many cycles.

- Water harvesting systems based on absorption are the best and most popular option for use when water is needed in environments with low humidity, with an increased focus on fabricated materials that have a high ability to absorb water and release it at the lowest time and cost to achieve the highest quantity of water harvesting cycles. Furthermore, it is recommended that the feasibility of designing combined fog harvesting meshes with absorption-based systems be investigated in order to increase the system's water productivity.
- When small amounts of water are needed and a clean or low-cost energy source is available, TEC may be used. It may also be used to collect a small amount of water to enhance the cooling process of specific equipment.
- VCRC water harvesting systems provide large water when compared to other types, but they require and consume a lot of energy if it isn't designed efficiently. As a result, it is advised to implement a strategy aimed at designing and establishing water harvesting systems while minimizing energy consumption.
- Since there are numerous regions around the world where biomass is burned and gas is produced, it is suggested here to power active water harvesting systems with the harvested energy.
- Using the air conditioner for water harvesting and air conditioning has been studied and indicated to be very valuable; therefore, it is suggested to investigate the feasibility of using the air conditioner primarily for harvesting atmospheric water at the lowest possible cost and considering it as a commercial water harvesting machine.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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